MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COMPONENTS, MISCELLANEOUS, KEY CODE FMD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic miscellaneous components module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.119 failures per million operating hours for class I or 0.167 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FMD.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 80 contact pins positioned A1 through A40 and B1 through B40.

3.5.5 Fin position. The module shall have an insertion and extraction fin on increment A and heat dissipation fins on increments A and B.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/310-1 for class I and M28787/310-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

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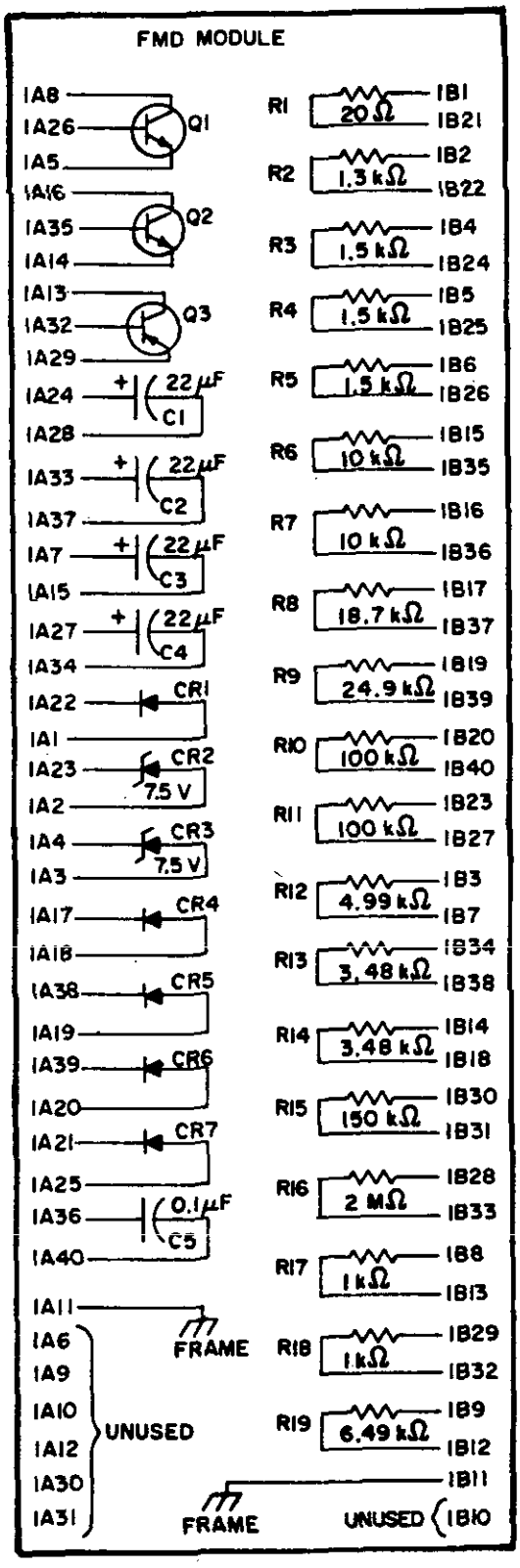


FIGURE 1. FMD module function diagram, miscellaneous components.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ 2/	25°C ±5°C 3/		Class I 3/				Class II 3/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1	Resistance													
	Initial	II	X	19.8	20.2	19.7	20.3	19.7	20.3	19.62	20.38	19.68	20.32	Ω
	End-of-life	II	X	19.7	20.2	19.6	20.4	19.6	20.4	19.59	20.48	19.52	20.48	Ω
A2	Resistance													
	Initial	II	X	1.287	1.313	1.281	1.320	1.281	1.320	1.275	1.325	1.279	1.321	kΩ
	End-of-life	II	X	1.281	1.320	1.274	1.326	1.274	1.326	1.269	1.331	1.269	1.331	kΩ
A3	Resistance													
	Initial	II	X	1.485	1.515	1.478	1.523	1.478	1.523	1.472	1.529	1.476	1.542	kΩ
	End-of-life	II	X	1.478	1.523	1.470	1.530	1.470	1.530	1.464	1.536	1.464	1.536	kΩ
A4, A5	Resistance													
	Initial	II	X	1.425	1.575	1.380	1.620	1.380	1.620	1.275	1.725	1.365	1.635	kΩ
	End of-life	II	X	1.335	1.665	1.290	1.710	1.290	1.710	1.185	1.815	1.245	1.755	kΩ
A6, A7	Resistance													
	Initial	II	X	9.90	10.10	9.85	10.15	9.85	10.15	9.81	10.19	9.84	10.16	kΩ
	End-of-life	II	X	9.85	10.15	9.80	10.20	9.80	10.20	9.76	10.24	9.76	10.24	kΩ
A8	Resistance													
	Initial	II	X	18.51	18.89	18.42	18.98	18.42	18.98	18.35	19.06	18.40	19.00	kΩ
	End-of-life	II	X	18.42	18.98	18.33	19.07	18.33	19.07	18.25	19.15	18.25	19.15	kΩ
A9	Resistance													
	Initial	II	X	24.65	25.15	24.53	25.27	24.53	25.27	24.42	25.37	24.50	25.29	kΩ
	End-of-life	II	X	24.53	25.27	24.40	25.40	24.40	25.40	24.30	25.50	24.30	25.50	kΩ
A10, A11	Resistance													
	Initial	II	X	99.0	101.0	98.5	101.5	98.5	101.5	98.1	101.9	98.4	101.6	kΩ
	End-of-life	II	X	98.5	101.5	98.0	102.0	98.0	102.0	97.6	102.4	97.6	102.4	kΩ
A12	Resistance													
	Initial	II	X	4.940	5.040	4.915	5.065	4.915	5.065	4.895	5.085	4.910	5.070	kΩ
	End-of-life	II	X	4.915	5.065	4.890	5.090	4.890	5.090	4.870	5.110	4.870	5.110	kΩ

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 2/	25°C ±5°C 3/		Class I 3/				Class II 3/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A13, A14	Resistance													
	Initial	II	X	3.445	3.515	3.428	3.532	3.428	3.532	3.414	3.546	3.424	3.536	kΩ
	End-of-life	II	X	3.428	3.532	3.410	3.550	3.410	3.550	3.396	3.564	3.396	3.564	kΩ
A15	Resistance													
	Initial	II	X	148.5	151.5	147.8	152.3	147.8	152.3	147.2	152.9	147.7	152.4	kΩ
	End-of-life	II	X	147.8	152.3	147.0	153.0	147.0	153.0	146.4	153.6	146.4	153.6	kΩ
A16	Resistance													
	Initial	II	X	1.9	2.1	1.76	2.24	1.76	2.24	1.50	2.50	1.70	2.30	MΩ
	End-of-life	II	X	1.78	2.22	1.64	2.36	1.64	2.36	1.38	2.62	1.48	2.52	MΩ
A17, A18	Resistance													
	Initial	II	X	0.990	1.010	0.985	1.015	0.985	1.015	0.981	1.019	0.984	1.016	kΩ
	End-of-life	II	X	0.985	1.015	0.980	1.020	0.980	1.020	0.976	1.024	0.976	1.024	kΩ
A19	Resistance													
	Initial	II	X	6.425	6.555	6.393	6.587	6.393	6.587	6.367	6.613	6.386	6.594	kΩ
	End-of-life	II	X	6.393	6.587	6.360	6.620	6.360	6.620	6.334	6.646	6.334	6.646	kΩ
B1-B4	Capacitance													
	Initial	III	X	19.8	24.2	18.7	25.3	18.7	25.3	17.6	26.4	17.6	26.4	µF
	End-of-life	III	X	18.7	25.3	17.6	26.4	17.6	26.4	16.5	27.5	16.5	27.5	µF
B5	Capacitance													
	Initial	III	X	0.090	0.110	0.080	0.120	0.080	0.120	0.075	0.125	0.075	0.125	µF
	End-of-life	III	X	0.080	0.120	0.075	0.125	0.075	0.125	0.070	0.130	0.070	0.130	µF
C1-C4	Dissipation													
	Initial	III	X	0	6	0	6	0	6	0	6	0	6	%
	End-of-life	III	X	0.0	7.5	0.0	7.5	0.0	7.5	0.0	7.5	0.0	7.5	%
C5	Dissipation													
	Initial	III	X	0.0	2.5	0.00	3.25	0.0	2.5	0.0	4.0	0.0	2.5	%
	End-of-life	III	X	0.0	4.0	0.00	4.75	0.0	4.0	0.0	5.5	0.0	4.0	%
D1-D4	DC leakage current	IV	X	0	2	0	2	0	2	0	2	0	2	µA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 2/	25°C ±5°C 3/		Class I 3/				Class II 3/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
E1	Forward voltage drop (V_F), I_F = 1 A													
	Initial	V	X	0.7	1.2	0.75	1.25	0.63	1.13	0.86	1.36	0.55	1.05	V
	End-of-life	V	X	0.65	1.25	0.70	1.30	0.58	1.18	0.81	1.41	0.50	1.10	V
E2, E3	Forward voltage drop (V_F), I_F = 200 mA													
	Initial	V	X	0.0	1.5	0.00	1.55	0.00	1.43	0.00	1.66	0.00	1.35	V
	End-of-life	V	X	0.00	1.55	0.0	1.6	0.00	1.48	0.00	1.71	0.0	1.4	V
E4-E7	Forward voltage drop (V_F), I_F = 10 mA													
	Initial	V	X	0	1	0.00	1.05	0.00	0.93	0.00	1.16	0.00	0.85	V
	End-of-life	V	X	0.00	1.05	0.0	1.1	0.00	0.98	0.00	1.21	0.0	0.9	V
F1	Reverse current (I_R), V_R = 200 V													
	Initial	VI	X	0	50	0	50	0	150	0	50	0	1000	nA
	End-of-life	VI	X	0	150	0	150	0	500	0	150	0	4000	nA
F2-F5	Reverse current (I_R), V_R = 20 V													
	Initial	VI	X	0	25	0	25	0	250	0	25	0	1000	nA
	End-of-life	VI	X	0	250	0	250	0	1000	0	250	0	4000	nA
G1, G2	Zener voltage (V_Z), I_Z = 250 µA	VII	X	6.60	7.87	6.60	7.87	6.60	8.00	6.60	7.87	6.60	8.10	V
G3, G4	Zener voltage (V_Z), I_Z = 20 mA	VII	X	7.13	7.87	7.00	7.87	7.13	8.00	6.80	7.87	7.13	8.10	V
G5, G6	Zener voltage (V_Z), I_Z = 50 mA	VII	X	7.13	8.00	7.00	8.00	7.13	8.15	6.80	8.00	7.13	8.30	V
H1, H2	Zener regulation check (ΔV_Z)	VII	X	-0.4	0.4	-0.4	0.4	-0.4	0.4	-0.4	0.4	-0.4	0.4	V
J	Reverse recovery time (t_{RR})	VIII	S	0	5	0	5	0	5	0	5	0	5	µs
K1-K3	Forward current transfer ratio (DC) check (h_{FE}), I_C = 1 mA	IX	X	75	∞	60	∞	75	∞	40	∞	75	∞	mA/ mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 2/	25°C ±5°C 3/		Class I 3/				Class II 3/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K4-K6	Forward current transfer ratio (DC) check (h_{FE}), I_C = 10 mA	IX	X	100	∞	85	∞	100	∞	75	∞	100	∞	mA/ mA
L1, L2	Forward current transfer ratio (pulsed) check (h_{FE}), I_C = 500 mA	X	X	30	∞	30	∞	30	∞	30	∞	30	∞	mA/ mA
L3	Forward current transfer ratio (pulsed) check (h_{FE}), I_C = -500 mA	X	X	50	∞	50	∞	50	∞	50	∞	50	∞	mA/ mA
M1, M2	Saturation voltage (V_{CE}(SAT)), I_C = 150 mA, I_B = 15 mA	XI	X	0	300	0	300	0	300	0	300	0	300	mV
M3	Saturation voltage (V_{CE}(SAT)), I_C = -150 mA, I_B = -15 mA	XI	X	0	400	0	400	0	400	0	400	0	400	mV
M4, M5	Saturation voltage (V_{CE}(SAT)), I_C = 500 mA, I_C = 50 mA	XI	X	0	1	0	1	0	1	0	1	0	1	V
M6	Saturation voltage (V_{CE}(SAT)), I_C = -500 mA, I_B = -50 mA	XI	X	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	V
N1, N2	NPN turn-on time (t_{ON})	XII	S	0	45	0	45	0	50	0	45	0	55	ns
N3	PNP turn-on time (t_{ON})	XIII	S	0	55	0	55	0	60	0	55	0	65	ns
P1, P2	NPN turn-off time (t_{OFF})	XIV	S	0	300	0	300	0	350	0	300	0	400	ns
P3	PNP turn-off time (t_{OFF})	XV	S	0	300	0	300	0	350	0	300	0	400	ns
Q1, Q2	Breakdown voltage check (emitter-to-base)	XVI	S	6	---	6	---	6	---	6	---	6	---	V
Q3	Breakdown voltage check (emitter-to-base)	XVI	S	---	-5	---	-5	---	-5	---	-5	---	-5	V
R1, R2	Breakdown voltage check (collector-to-emitter)	XVII	S	50	---	50	---	50	---	50	---	50	---	V
R3	Breakdown voltage check (collector-to-emitter)	XVII	S	---	-60	---	-60	---	-60	---	-60	---	-60	V

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 2/	25°C ±5°C 3/		Class I 3/				Class II 3/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S1-S3	Isolation													
	Initial	XVIII	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	XVIII	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
T1, T2	Continuity	XIX	S	0	1	0	1	0	1	0	1	0	1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ Unless otherwise specified, limits are for initial and end-of-life.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. If test conditions Q, R, or S are part of a sequence of tests, the order of testing shall be as follows: Q, R, S where applicable, followed by the remaining tests in the sequence. If Q, R, or S are performed at any time during the quality assurance inspection testing, these shall be followed by, at a minimum, performance of all 100 percent tests.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test conditions S and T, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A through H and K through M shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through T shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Power dissipation	4 max	4 max	W

6.5 Module function. This module contains 19 resistors, 5 capacitors, 5 diodes, 2 zener diodes, 1 PNP transistor, and 2 NPN transistors to be used for programming operational amplifier modules.

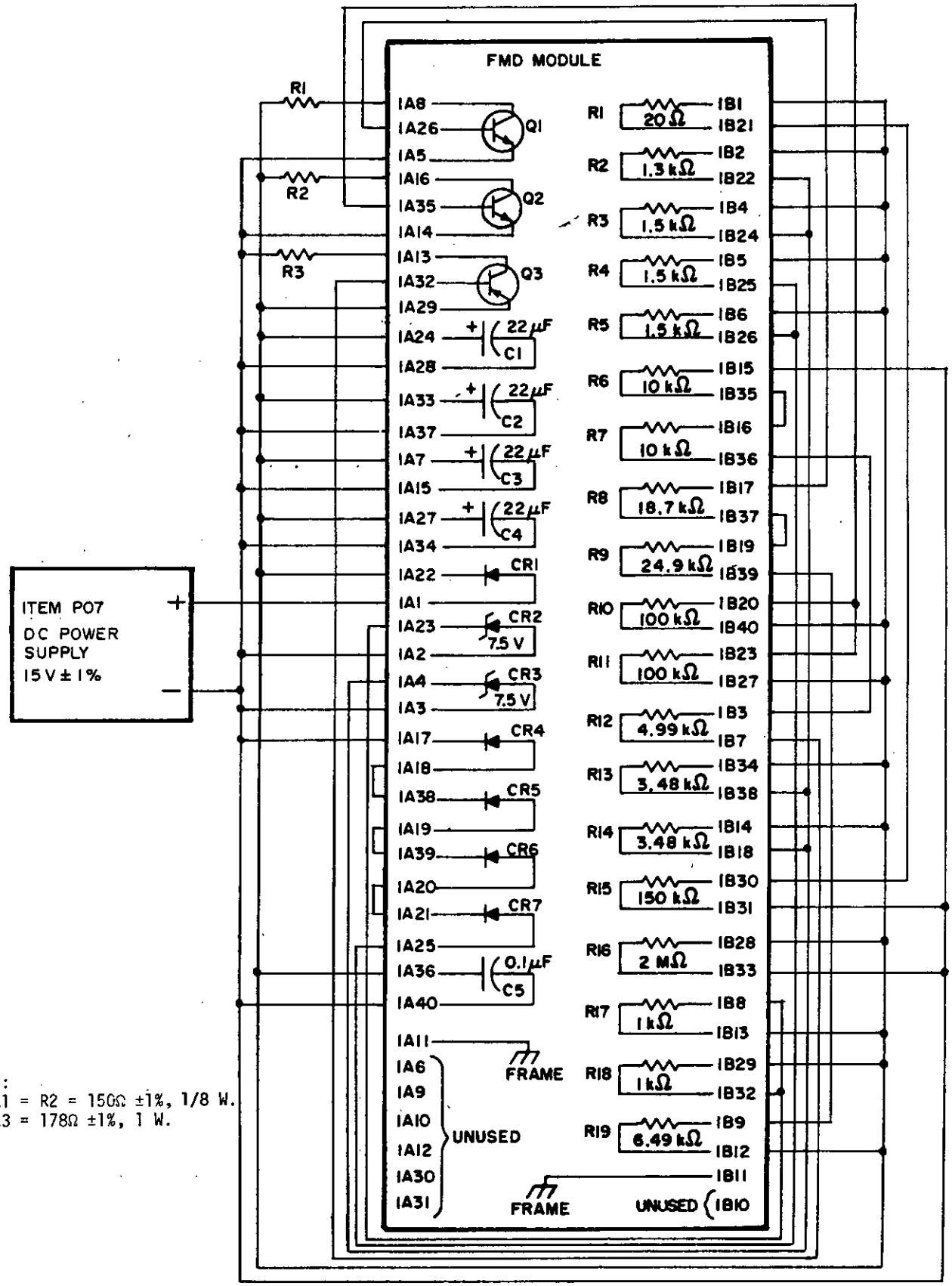
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES:
1. R1 = R2 = 150Ω ±1%, 1/8 W.
2. R3 = 178Ω ±1%, 1 W.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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TABLE II. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	R1 Resistance	1B1	1B21	19.8	20.2	Ω	3
A2	R2 "	1B2	1B22	1.287	1.313	kΩ	"
A3	R3 "	1B4	1B24	1.485	1.515	"	"
A4	R4 "	1B5	1B25	1.425	1.575	"	"
A5	R5 "	1B6	1B26	1.425	1.575	"	"
A6	R6 "	1B15	1B35	9.90	10.10	"	"
A7	R7 "	1B16	1B36	9.90	10.10	"	"
A8	R8 "	1B17	1B37	18.51	18.89	"	"
A9	R9 "	1B19	1B39	24.65	25.15	"	"
A10	R10 "	1B20	1B40	99.0	101.0	"	"
A11	R11 "	1B23	1B27	99.0	101.0	"	"
A12	R12 "	1B3	1B7	4.940	5.040	"	"
A13	R13 "	1B34	1B38	3.445	3.515	"	"
A14	R14 "	1B14	1B18	3.445	3.515	"	"
A15	R15 "	1B30	1B31	148.5	151.5	"	"
A16	R16 "	1B28	1B33	1.9	2.1	MΩ	"
A17	R17 "	1B8	1B13	0.990	1.010	kΩ	"
A18	R18 "	1B29	1B32	0.990	1.010	"	"
A19	R19 "	1B9	1B12	6.425	6.555	"	"

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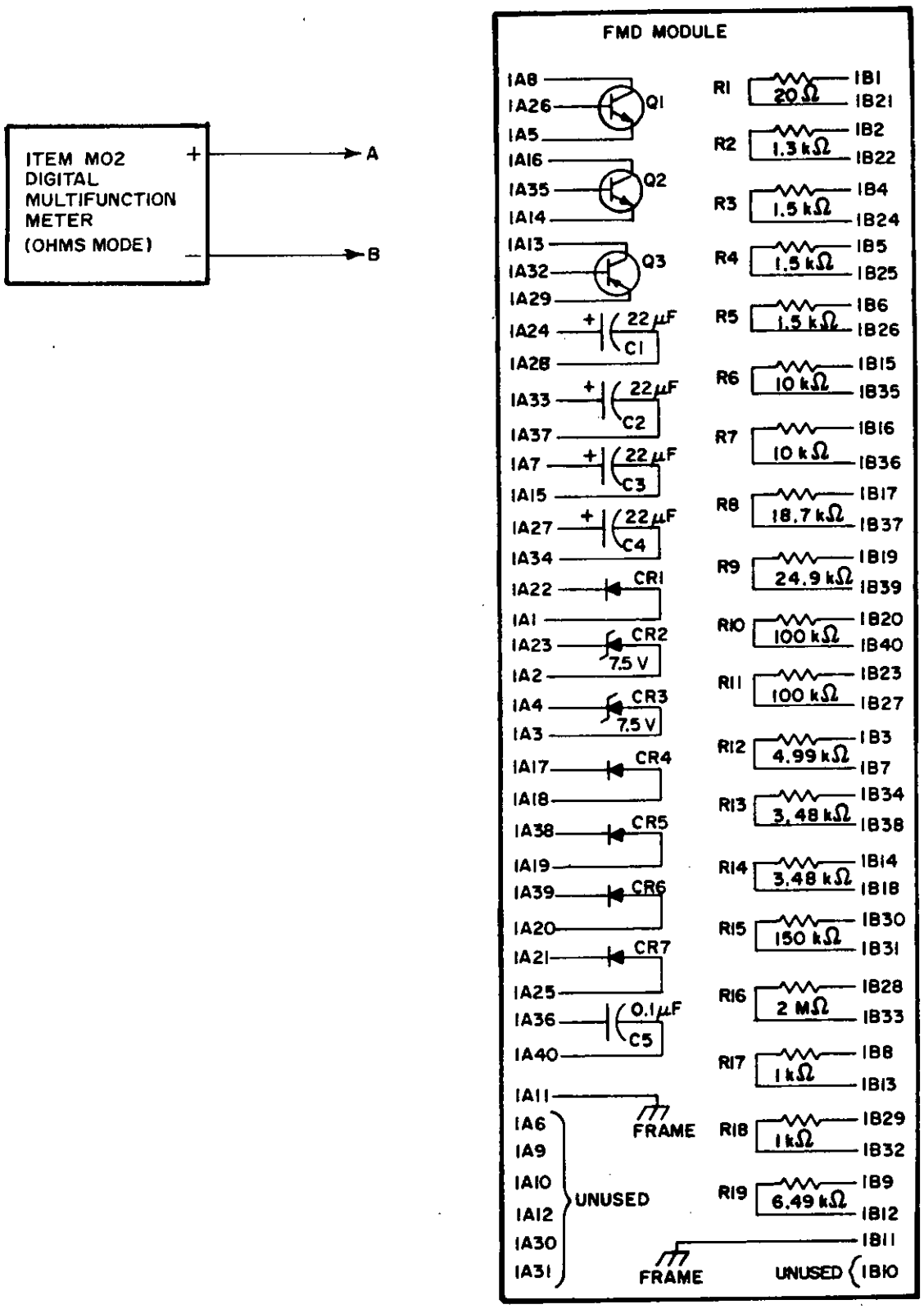


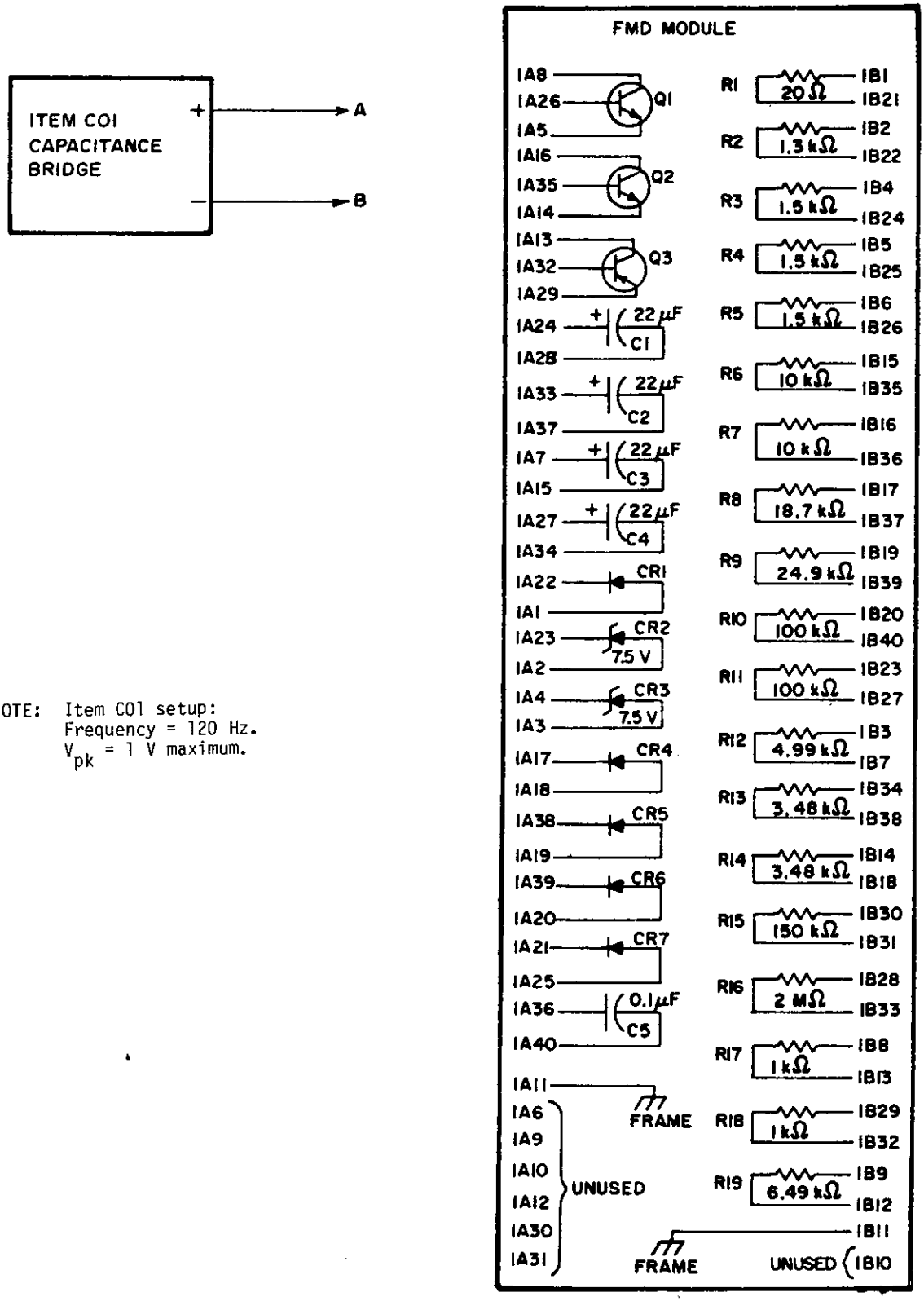
FIGURE 3. Resistance test circuit for table II.

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TABLE III. Capacitance and dissipation test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	C1	Capacitance	1A24	1A28	19.8	24.2	µF	4
B2	C2	"	1A33	1A37	"	"	"	"
B3	C3	"	1A7	1A15	"	"	"	"
B4	C4	"	1A27	1A34	"	"	"	"
B5	C5	"	1A36	1A40	0.090	0.110	"	"
C1	C1	Dissipation	1A24	1A28	0	6	%	"
C2	C2	"	1A33	1A37	"	"	"	"
C3	C3	"	1A7	1A15	"	"	"	"
C4	C4	"	1A27	1A34	"	"	"	"
C5	C5	"	1A36	1A40	0.0	2.5	"	"

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NOTE: Item C01 setup:
Frequency = 120 Hz.
V_{pk} = 1 V maximum.

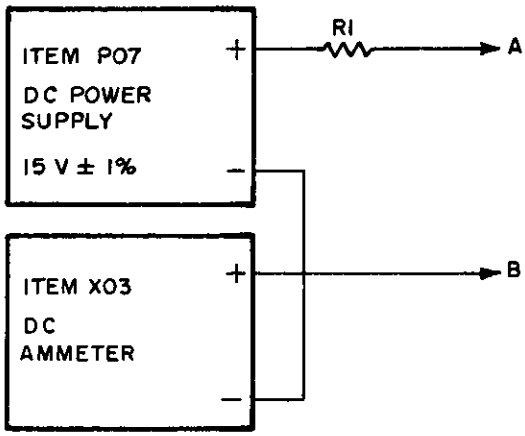
FIGURE 4. Capacitance and dissipation test circuit for table III.

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TABLE IV. DC leakage current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D1	C1	DC leakage current	1A24	1A28	0	2	µA	5
D2	C2	"	1A33	1A37	"	"	"	"
D3	C3	"	1A7	1A15	"	"	"	"
D4	C4	"	1A27	1A34	"	"	"	"

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NOTES:
1. R1 = 1 kΩ ±1%, 1/8 W.
2. Unit conforms to requirements if leakage drops below limit indicated in table IV in 5 minutes or less

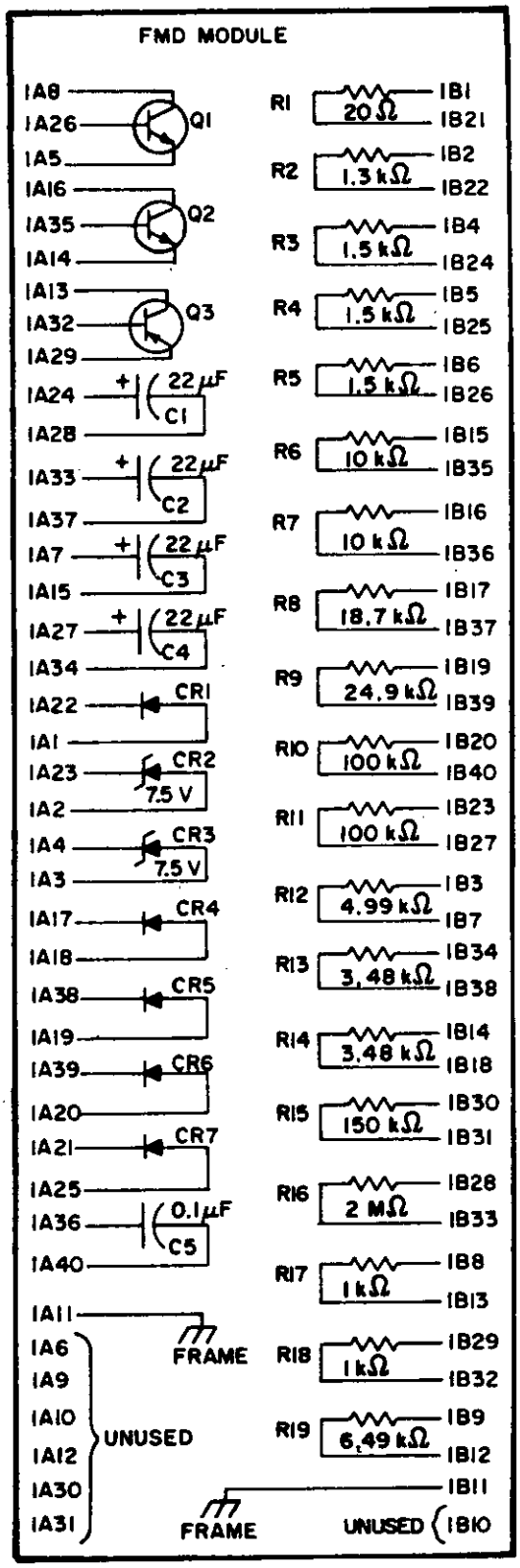


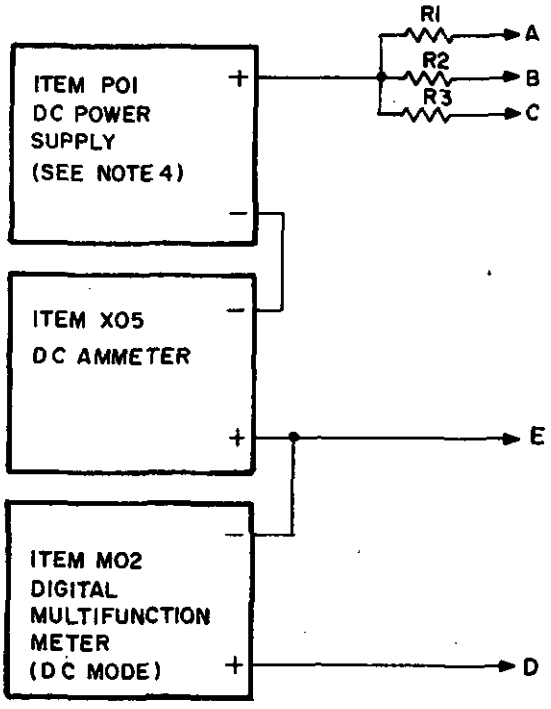
FIGURE 5. DC leakage current test circuit for table IV.

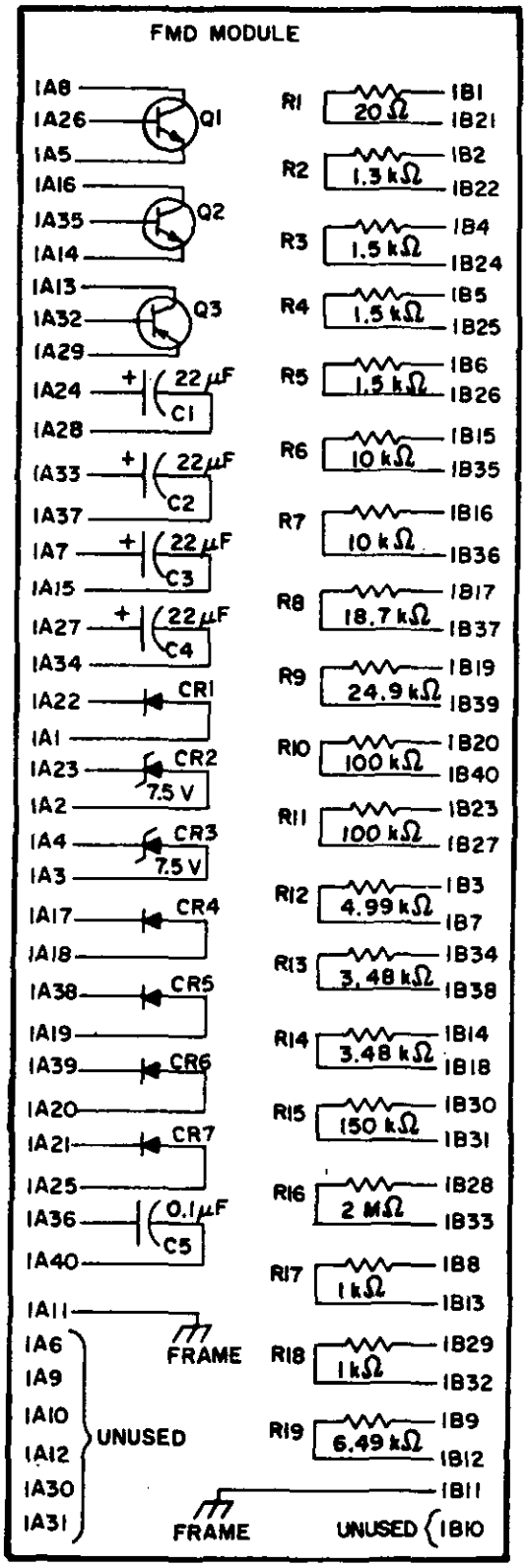
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* TABLE V. Forward voltage drop (V_F) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	Forward voltage drop (V_F), I_F = 1 A	1A1	Open	Open	1A1	1A22	0.7	1.2	V	6
E2	Forward voltage drop (V_F), I_F = 200 mA	Open	1A2	"	1A2	1A23	0.0	1.5	"	"
E3	Forward voltage drop (V_F), I_F = 200 mA	"	1A3	"	1A3	1A4	0.0	1.5	"	"
E4	Forward voltage drop (V_F), I_F = 10 mA	"	Open	1A18	1A18	1A17	0	1	"	"
E5	"	"	"	1A19	1A19	1A38	"	"	"	"
E6	"	"	"	1A20	1A20	1A39	"	"	"	"
E7	"	"	"	1A25	1A25	1A21	"	"	"	"

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NOTES:
1. R1 = 10Ω ±1%, 20 W.
2. R2 = 100Ω ±1%, 5 W.
3. R3 = 1 kΩ ±1%, 1/8 W.
4. a. For test condition E1, adjust item P01 to obtain a forward current of 1 A ±5%.
   b. For test condition E2 and E3, adjust item P01 to obtain a forward current of 200 mA ±5%.
   c. For test conditions E4-E7, adjust item P01 to obtain a forward current of 10 mA ±5%.

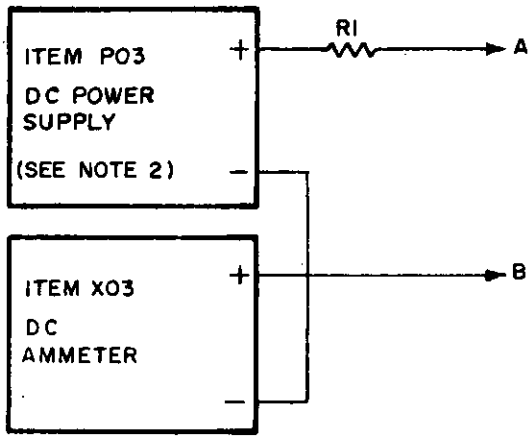
* FIGURE 6. Forward voltage drop (V_F) test circuit for table V.

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TABLE VI. Reverse current (I_R) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	CR1 Reverse current (I_R), V_R = 200 V	1A22	1A1	0	50	nA	7
F2	CR4 Reverse current (I_R), V_R = 20 V	1A17	1A18	"	25	"	"
F3	CR5 "	1A38	1A19	"	"	"	"
F4	CR6 "	1A39	1A20	"	"	"	"
F5	CR7 "	1A21	1A25	"	"	"	"

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NOTES:
1. R1 = 200 kΩ ±1%, 1/8 W.
2. a. For test condition F1, adjust item PO3 for 200 V ±1%.
   b. For test conditions F2-F5, adjust item PO3 for 20 V ±1%.

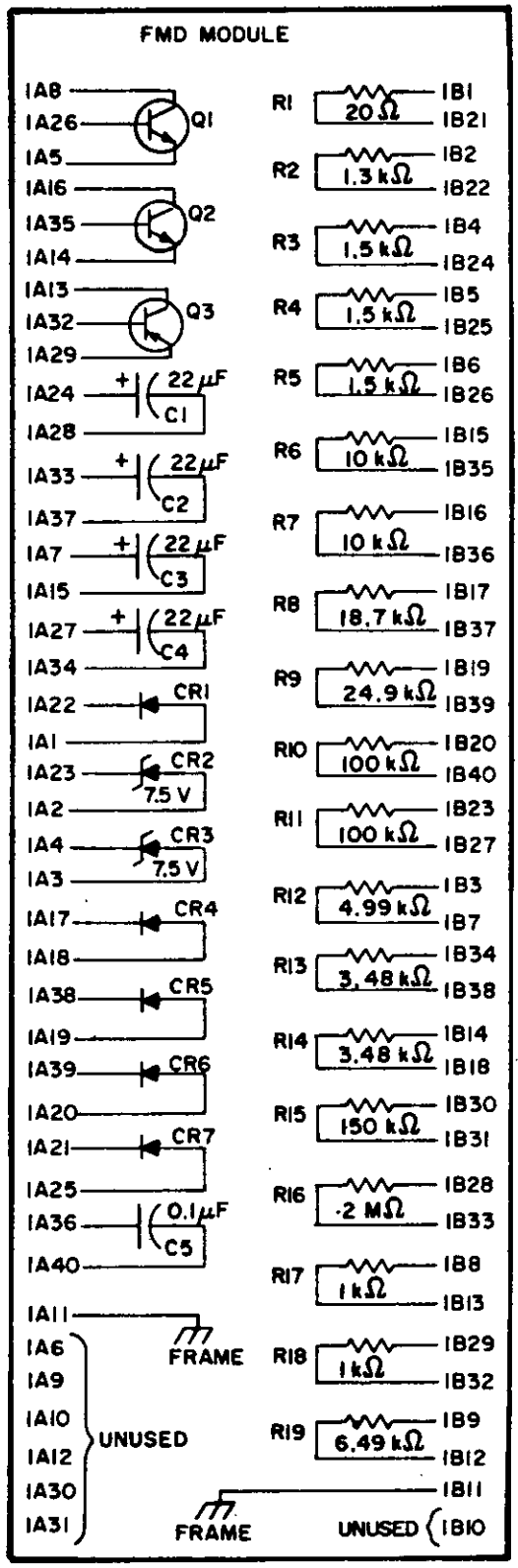


FIGURE 7. Reverse current (I_R) test circuit for table VI.

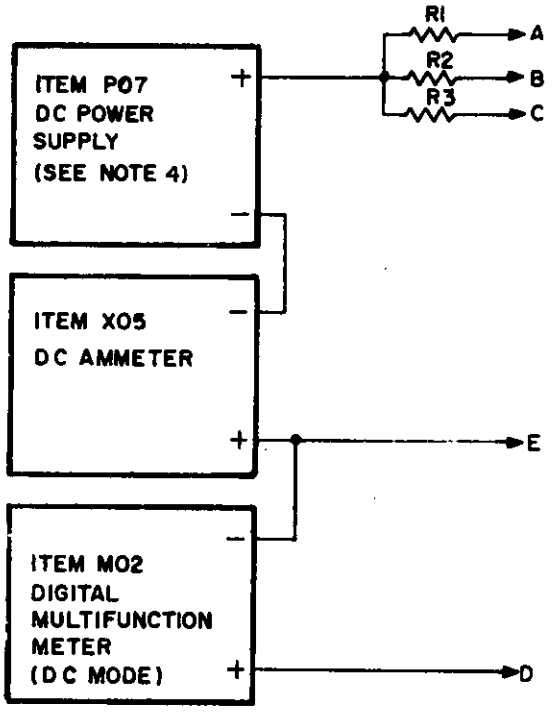
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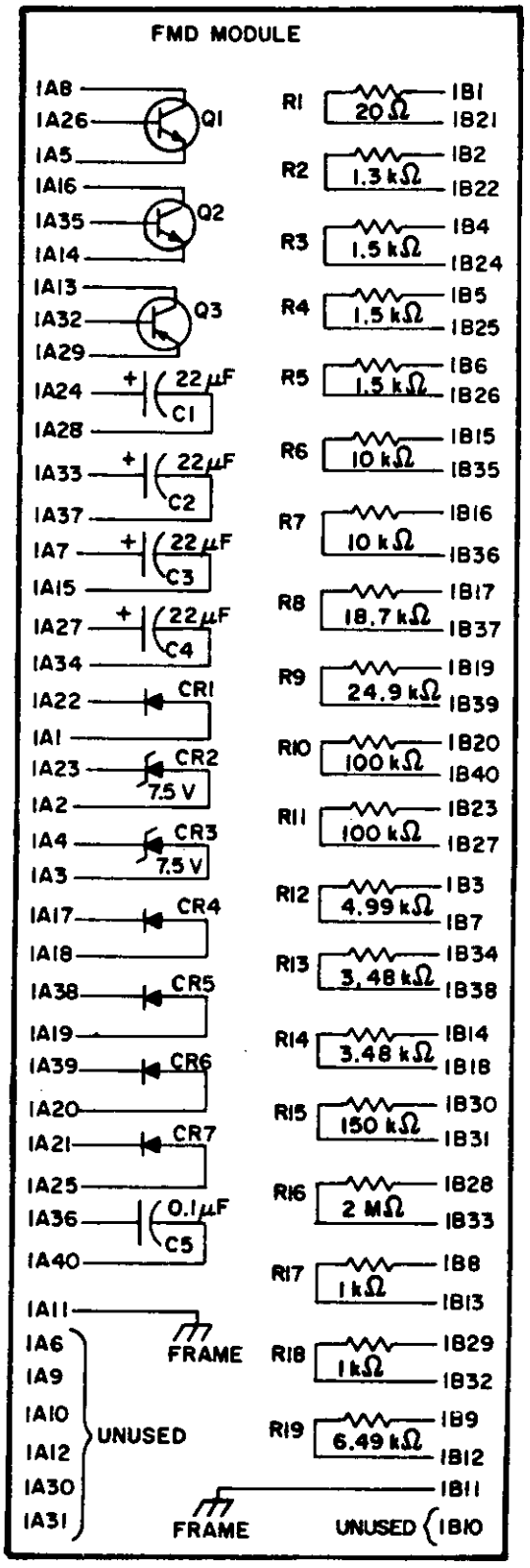
TABLE VII. Zener voltage (V_Z) and zener regulation check (ΔV_Z) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G1	CR2 Zener voltage (V_Z), I_Z = 250 µA	1A23	Open	Open	1A23	1A2	6.60	7.87	V	8
G2	CR3 Zener voltage (V_Z), I_Z = 250 µA	1A4	Open	"	1A4	1A3	6.60	"	"	"
G3	CR2 Zener voltage (V_Z), I_Z = 20 mA	Open	1A23	"	1A23	1A2	7.13	"	"	"
G4	CR3 Zener voltage (V_Z), I_Z = 20 mA	"	1A4	"	1A4	1A3	"	"	"	"
G5	CR2 Zener voltage (V_Z), I_Z = 50 mA	"	Open	1A23	1A23	1A2	"	8.00	"	"
G6	CR3 Zener voltage (V_Z), I_Z = 50 mA	"	Open	1A4	1A4	1A3	"	8.00	"	"
H1	Zener regulation check (ΔV_Z) 1/	---	---	---	---	---	-0.4	0.4	"	"
H2	Zener regulation check (ΔV_Z) 2/	---	---	---	---	---	-0.4	0.4	"	"

1/ Test condition H1 = test condition G3 - test condition G1.
2/ Test condition H2 = test condition G4 - test condition G2.

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NOTES:
1. R1 = 20 kΩ ±1%, 1/8 W.
2. R2 = 200Ω ±1%, 1/8 W.
3. R3 = 100Ω ±1%, 1/2 W.
4. a. For test conditions G1 and G2, adjust item P07 to obtain a current of 250 µA ±1%.
   b. For test conditions G3 and G4 adjust item P07 to obtain a current of 20 mA ±1%.
   c. For test conditions G5 and G6, adjust item P07 to obtain a current of 50 mA ±1%.
5. Zener breakdown voltage is read on item M02 after the specified test current has been applied for 20 ±2 seconds.

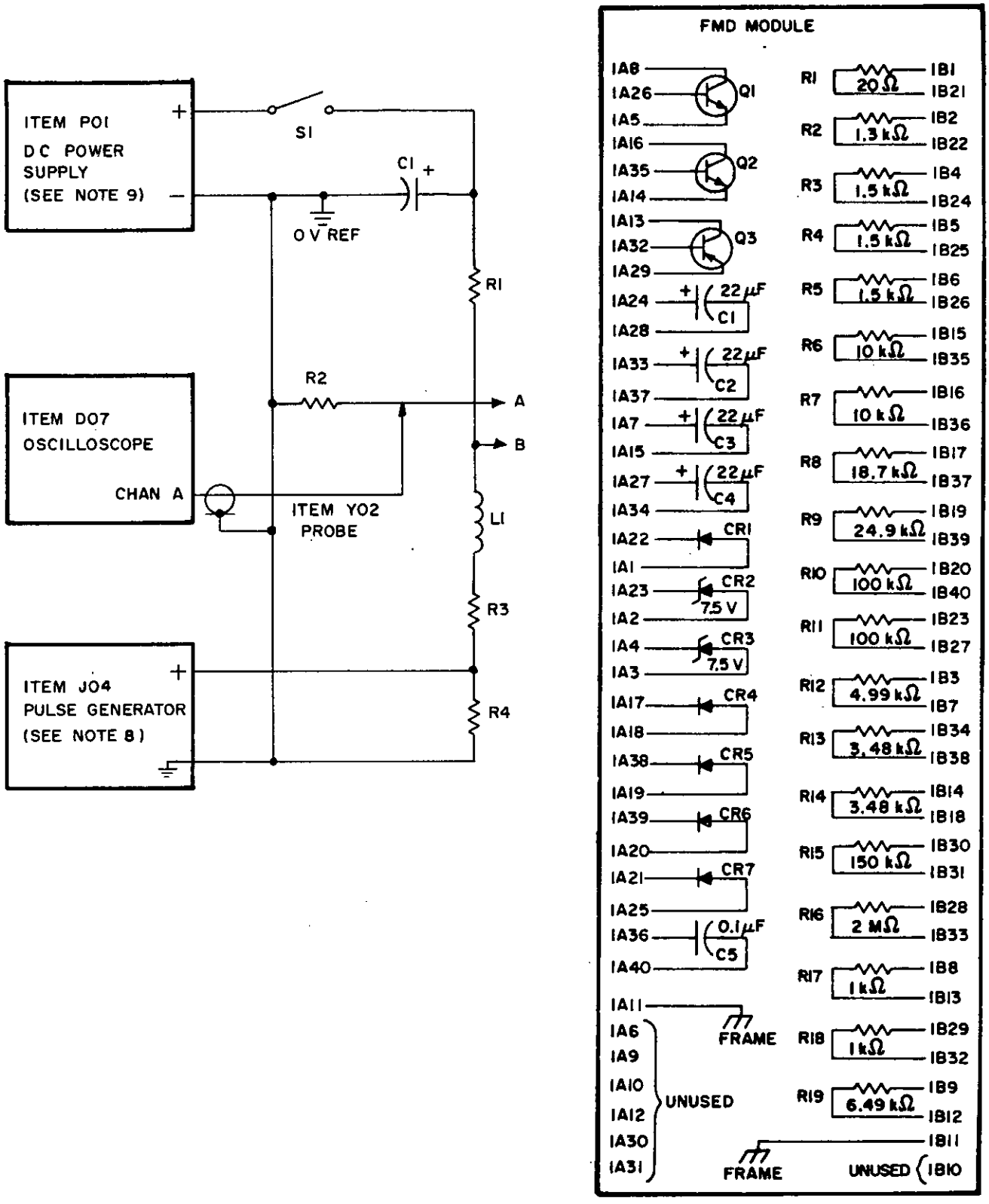
* FIGURE 8. Zener voltage (V_Z) and zener regulation check (ΔV_Z) test circuit for table VII.

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TABLE VIII. Reverse recovery time (t_{RR}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J	Reverse recovery time (t_{RR})	1A22	1A1	0	5	µs	9

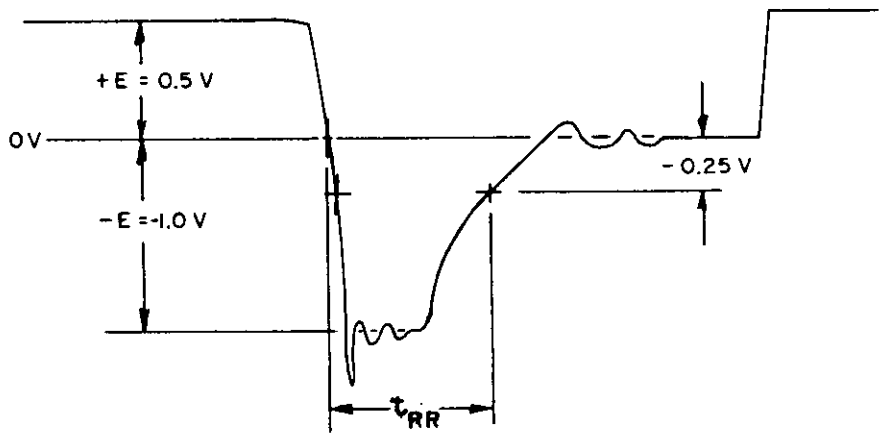
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* FIGURE 9. Reverse recovery time (t_{RR}) test circuit for table VIII.

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NOTES:
1. R1 = 30Ω ±5%, 50 W, noninductive.
2. R2 = 1Ω ±1%, 1 W, noninductive.
3. R3 = 11Ω ±10%, 1 W, noninductive.
4. R4 = 100Ω ±10%, 1 W, noninudctive.
5. C1 = 500 µF +100%, -10%, 50 V, electrolytic.
6. L1 = 1.0 ±0.5 µH (approximately 5 turns #14 or #16 wire).
7. S1 = Single pole, momentary, normally open, switch.
8. Item J04 pulse parameters:
   PRR = 1 ms ±5%.
   PW = 10 µs ±5%.
   Pulse polarity = Negative with baseline at 0 V.
   $t_{TLH} = t_{THL}$ = 15 ns maximum.
9. The test shall be performed as follows:
   a. Initially set item P01 and item J04 amplitudes to zero.
   b. Connect points A and B per table VIII.
   c. Close S1 and adjust item P01 for a +E of 0.5 V ±5% on item D07 (-E will not appear on item D07 yet). This represents a diode forward current of 0.5 amp.
   d. Adjust the item J04 amplitude for a -E of -1.0 ±5% on item D07. This represents a diode reverse current of 1.0 amp.
   e. Take reading t_{RR} from the time the negative going pulse crosses 0 V to the time it recovers to -0.25 V ±5% and does not ring below -0.25 V again. Wait 5 +5, -0 sec before taking reading.



* FIGURE 9. Reverse recovery time (t_{RR}) test circuit for table VIII - Continued.

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TABLE IX. Forward current transfer ratio (DC) check (h_{FE}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
K1	Q1 Forward current transfer ratio (DC) check (h_{FE}), I_C = 1 mA	1A26	Open	1A5	1A8	1A5	0.0	13.3	µA	10
K2	Q2 Forward current transfer ratio (DC) check (h_{FE}), I_C = 1 mA	1A35	"	1A14	1A16	1A14	"	"	"	"
K3	Q3 Forward current transfer ratio (DC) check (h_{FE}), I_C = -1 mA	1A29	"	1A32	1A29	1A13	"	"	"	"
K4	Q1 Forward current transfer ratio (DC) check (h_{FE}), I_C = 10 mA	Open	1A26	1A5	1A8	1A5	0	100	"	"
K5	Q2 Forward current transfer ratio (DC) check (h_{FE}), I_C = 10 mA	"	1A35	1A14	1A16	1A14	"	"	"	"
K6	Q3 Forward current transfer ratio (DC) check (h_{FE}), I_C = -10 mA	"	1A29	1A32	1A29	1A13	"	"	"	"

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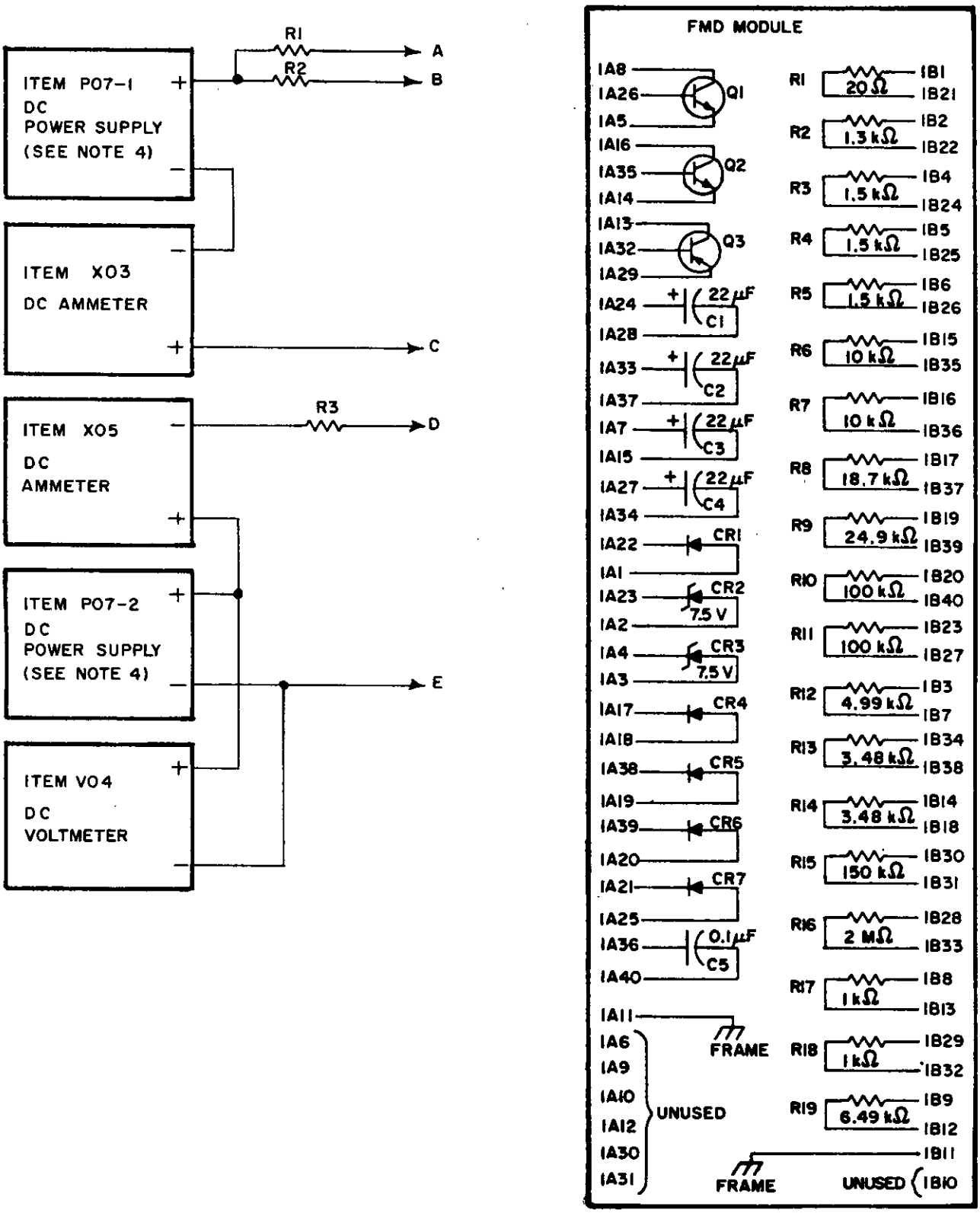


FIGURE 10. Forward current transfer ratio (DC) check (h_{FE}) test circuit for table IX.

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NOTES:
1. R1 = 1 MΩ ±1%, 1/8 W.
2. R2 = 100 kΩ ±1%, 1/8 W.
3. R3 = 10Ω ±1%, 1/4 W.
4. The test shall be performed as follows:
   a. Initially set items P07-1 and P07-2 to 0 V.
   b. Make the connections as per table IX.
   c. Adjust item P07-2 first for 10 V ±1% as measured on item V04.
   d. For test conditions K1-K3 adjust item P07-1 for a collector current of 1 ±1% mA as indicated on item X05.
   e. For test conditions K4-K6 adjust item P07-1 for a collector current of 10 mA ±1% as indicated on item X05.
   f. The currents measured on item X03 shall meet the limits of table IX.
5. Forward current transfer ratio (DC) = $\frac{1\ \text{mA}}{\text{Item X03 reading}}$ for K1-K3, and

   $= \frac{10\ \text{mA}}{\text{Item X03 reading}}$ for K4-K6
6. Item P07-1 and P07-2 must be isolated from chassis ground. Test leads shall be used which minimize stray capacitance to prevent transistor saturation.

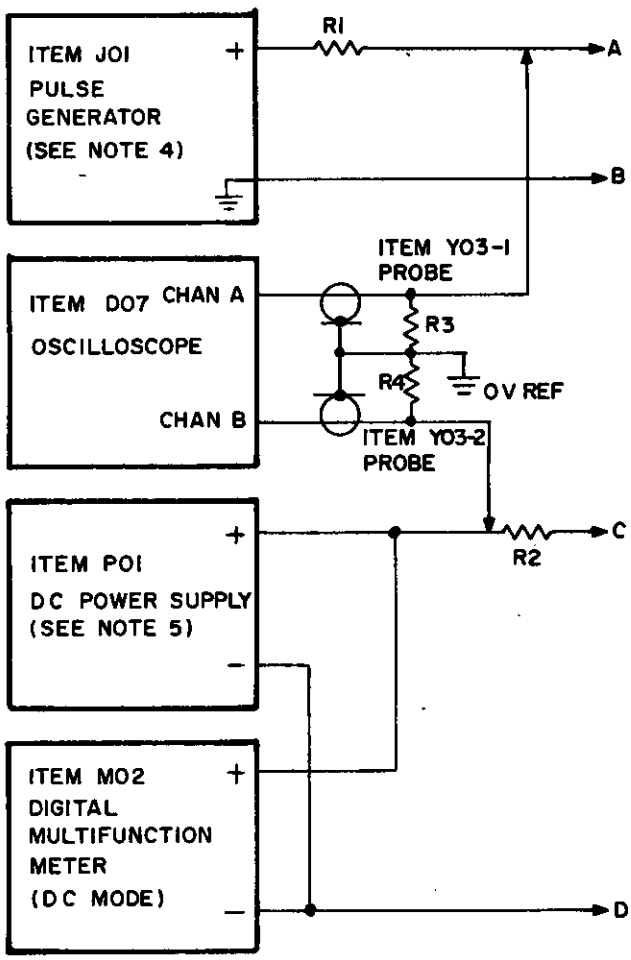
FIGURE 10. Forward current transfer ratio (DC) check (h_{FE}) test circuit for table IX - Continued.

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TABLE X. Forward current transfer ratio (pulsed) check (h_{FE}) test.

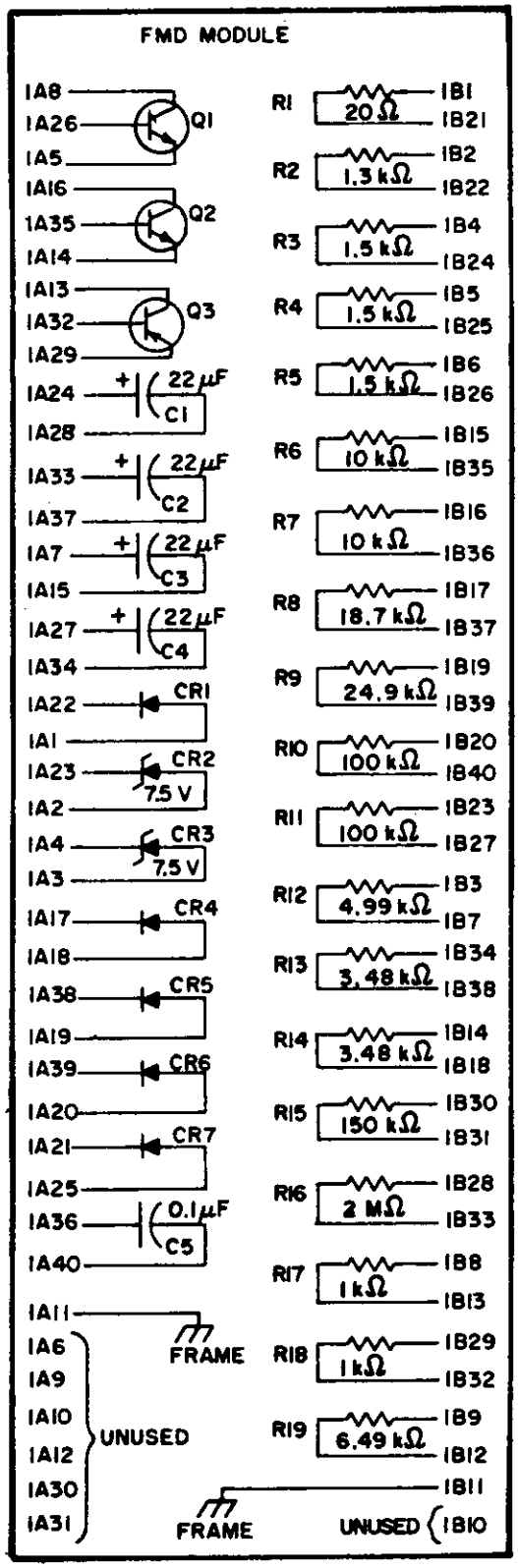
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	Q1 Forward current transfer ratio (pulsed) check (h_{FE}), I_C = 500 mA	1A26	1A5	1A8	1A5	0.00	16.67	mA	11
L2	Q2 Forward current transfer ratio (pulsed) check (h_{FE}), I_C = 500 mA	1A35	1A14	1A16	1A14	0.00	16.67	"	"
L3	Q3 Forward current transfer ratio (pulsed) check (h_{FE}), I_C = -500 mA	1A32	1A29	1A29	1A13	0	10	"	"

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CAUTION

Do not connect the negative terminals of items P01 or M02 to chassis ground.



* FIGURE 11. Forward current transfer ratio (pulsed) check (h_{FE}) test circuit for table X.

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NOTES:
1. R1 = 75Ω ±1%, 1/8 W.
2. R2 = 10Ω ±1%, 1/4 W.
3. R3 = R4 = 50Ω ±5%, 1 W.
4. Item J01 setup:
   PW = 20 ±2 µs.
   PRR = 1 kHz ±100 Hz.
   $t_{TLH} = t_{THL} \leq 50$ ns.
   For test conditions L1 and L2, set the polarity to positive with baseline at 0 V.
   For test condition L3 set the polarity to negative with baseline at 0 V.
5. The test shall be performed as follows:
   a. Initially set item P01 and item J01 amplitude to 0 V.
   b. Make the connections as per table X.
   c. Adjust item P01 for 10 V ±1% as indicated on item M02.
   d. Adjust item J01 amplitude to get a 0 to 500 mV positive pulse on channel B of item D07. This represents a collector current of 500 mA.
   e. Read the peak amplitude of the pulse on channel A in mV. This represents the base current in mA.
6. Forward current transfer ratio (pulsed) = $\frac{500 \text{ mA}}{|(\text{Item D07 reading})|}$

* FIGURE 11. Forward current transfer ratio (pulsed) check (h_{FE}) test circuit for table X - Continued.

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TABLE XI. Saturation voltage ($V_{CE}(SAT)$) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
M1	Q1	Saturation voltage ($V_{CE}(SAT)$), I_C = 150 mA, I_B = 15 mA	1A26	1A5	1A8	0	300	mV	12
M2	Q2	Saturation voltage ($V_{CE}(SAT)$), I_C = 150 mA, I_B = 15 mA	1A35	1A14	1A16	"	300	"	"
M3	Q3	Saturation voltage ($V_{CE}(SAT)$), I_C = -150 mA, I_B = -15 mA	1A32	1A13	1A29	"	400	"	"
M4	Q1	Saturation voltage ($V_{CE}(SAT)$), I_C = 500 mA, I_B = 50 mA	1A26	1A5	1A8	"	1	V	"
M5	Q2	Saturation voltage ($V_{CE}(SAT)$), I_C = 500 mA, I_B = 50 mA	1A35	1A14	1A16	"	1	"	"
M6	Q3	Saturation voltage ($V_{CE}(SAT)$), I_C = -500 mA, I_B = -50 mA	1A32	1A13	1A29	0.0	1.6	"	"

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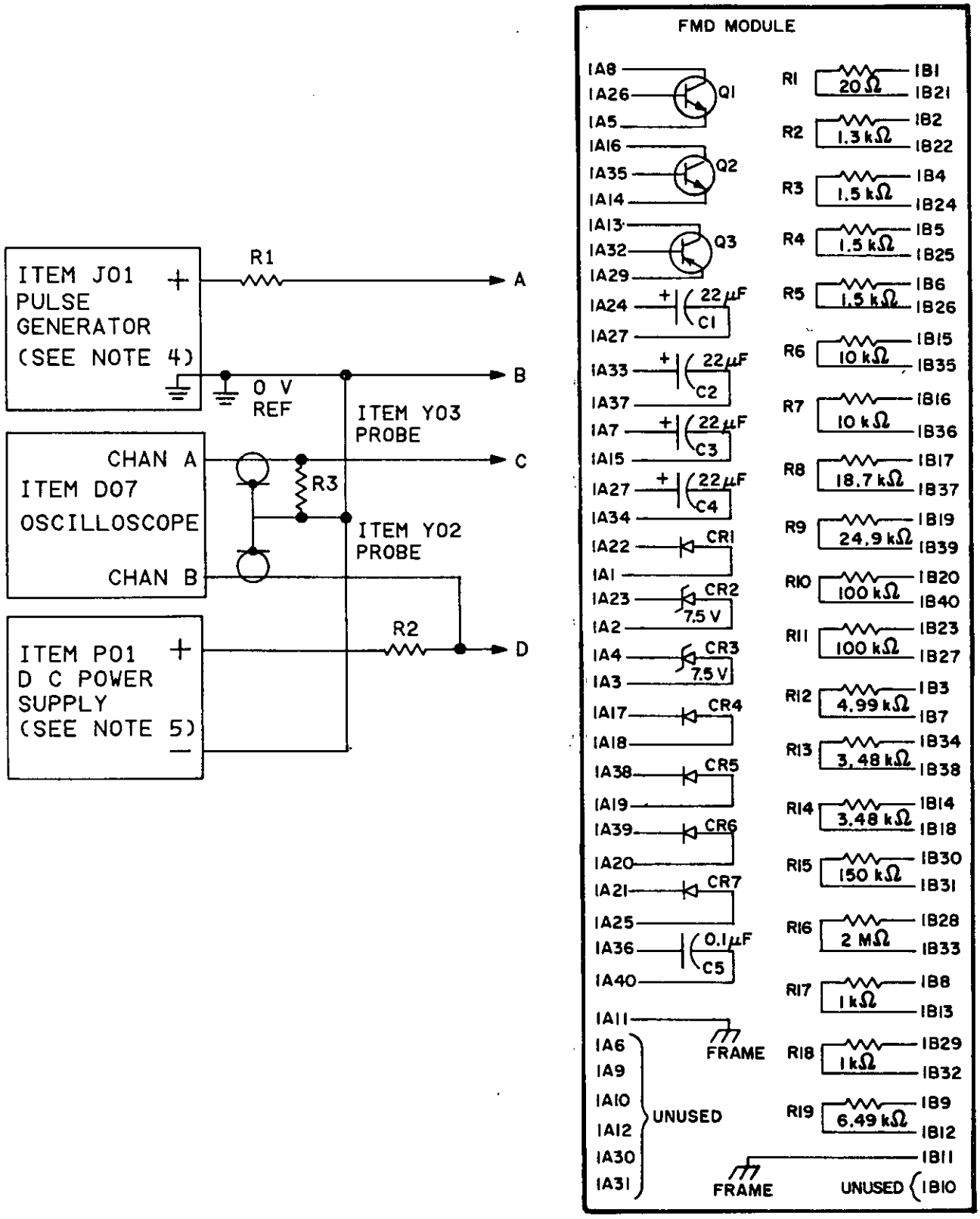
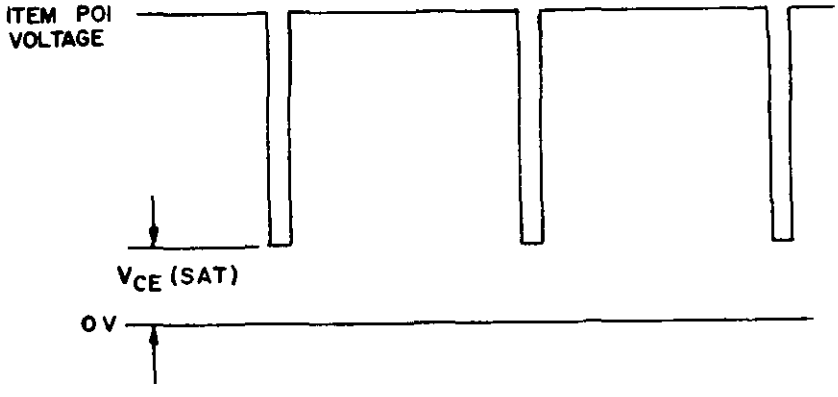


FIGURE 12. Saturation voltage (V_{CE}(SAT)) test circuit for table XI.

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NOTES:
1. R1 = 75Ω ±1%, 1/4 W.
2. R2 = 10Ω ±1%, 1/4 W.
3. R3 = 50Ω ±5%, 1 W.
4. Item J01 setup:
   PW = 20 ±2 µs.
   PRR = 1 kHz ±100 Hz.
   For test conditions M1, M2 and M4, M5, set the polarity to positive with base line at 0 V.
   For test conditions M3 and M6, set the polarity to negative with baseline at 6 V.
5. The test shall be performed as follows:
   a. Initially set item P01 and item J01 amplitude to 0 V.
   b. Make the connections as per table XI.
   c. For test conditions M1-M3, adjust item J01 amplitude to get a 0 to 15 mV pulse on channel A of item D07 with points A and D connected. This represents a pulsed base current of 15 mA. For test conditions M1 and M2, adjust item P01 to get a 0 to 150 mV positive pulse with points C and D connected. For test condition M3, connect points B and D. This represents a collector current of 150 mA. Read the collector saturation voltage on channel B of item D07.
   d. Repeat steps a through c for test conditions M4-M6 for a base current of 50 mA and a collector current of 500 mA.



* FIGURE 12. Saturation voltage (V_{CE}(SAT)) test circuit for table XI - Continued.

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*TABLE XII. NPN turn-on time (t_{ON}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Q1 NPN turn-on time (t_{ON})	1A26	1A8	1A5	0	45	ns	13,14
N2	Q2 NPN turn-on time (t_{ON})	1A35	1A16	1A14	0	45	ns	13,14

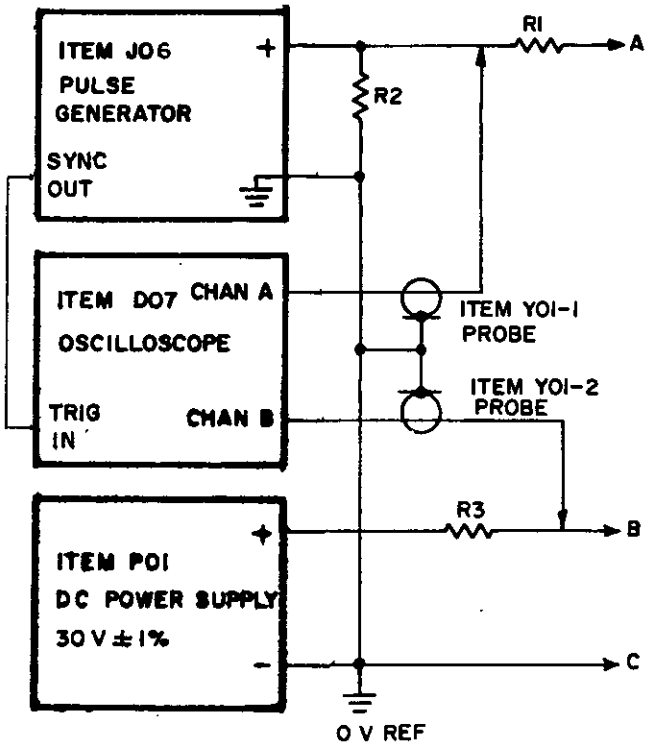
TABLE XIII. PNP turn-on time (t_{ON}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N3	Q3 PNP turn-on time (t_{ON})	1A32	1A29	1A13	0	55	ns	15,16

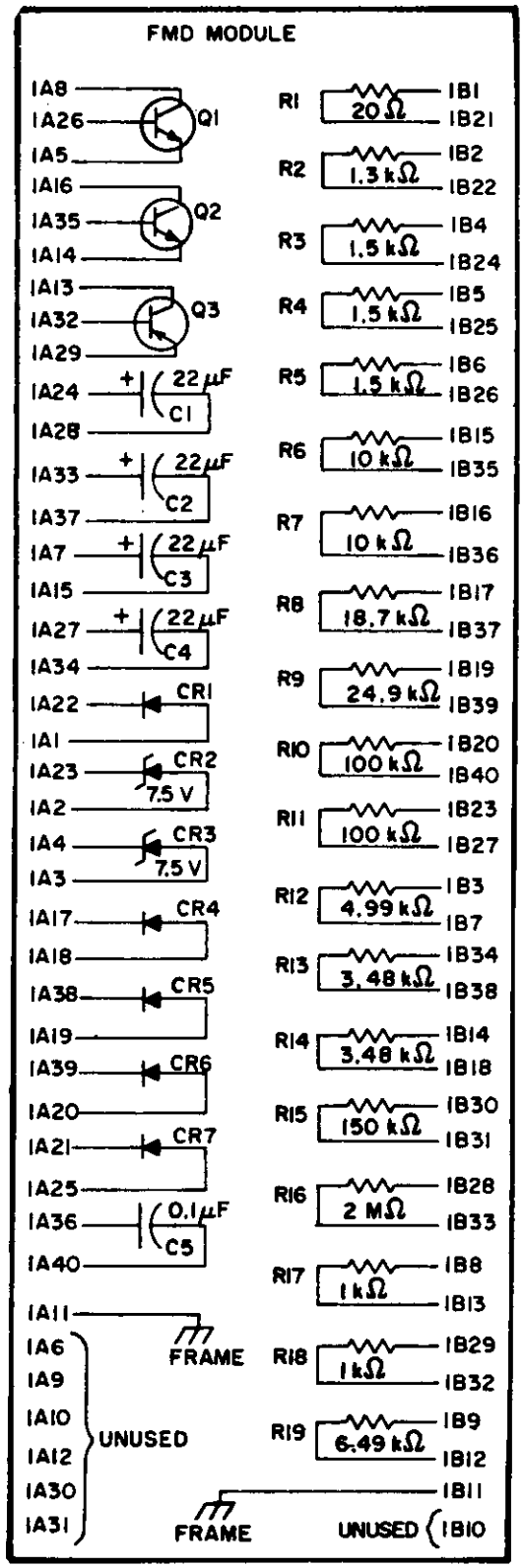
TABLE XIV. NPN turn-off time (t_{OFF}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Q1 NPN turn-off time (t_{OFF})	1A26	1A8	1A5	0	300	ns	17,18
P2	Q2 NPN turn-off time (t_{OFF})	1A35	1A16	1A14	0	300	ns	17,18

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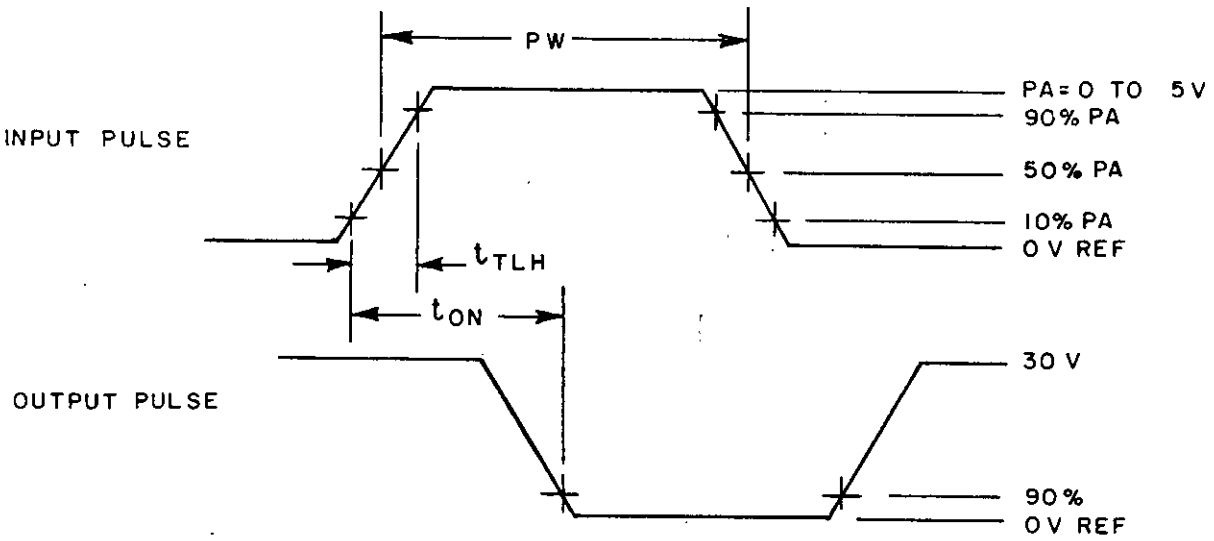


NOTES:
1. R1 = 200Ω ±1%, 1/4 W.
2. R2 = 50Ω ±5%, 1 W.
3. R3 = 200Ω ±1%, 1 W.



* FIGURE 13. NPN turn-on time (t_{ON}) test circuit for table XII.

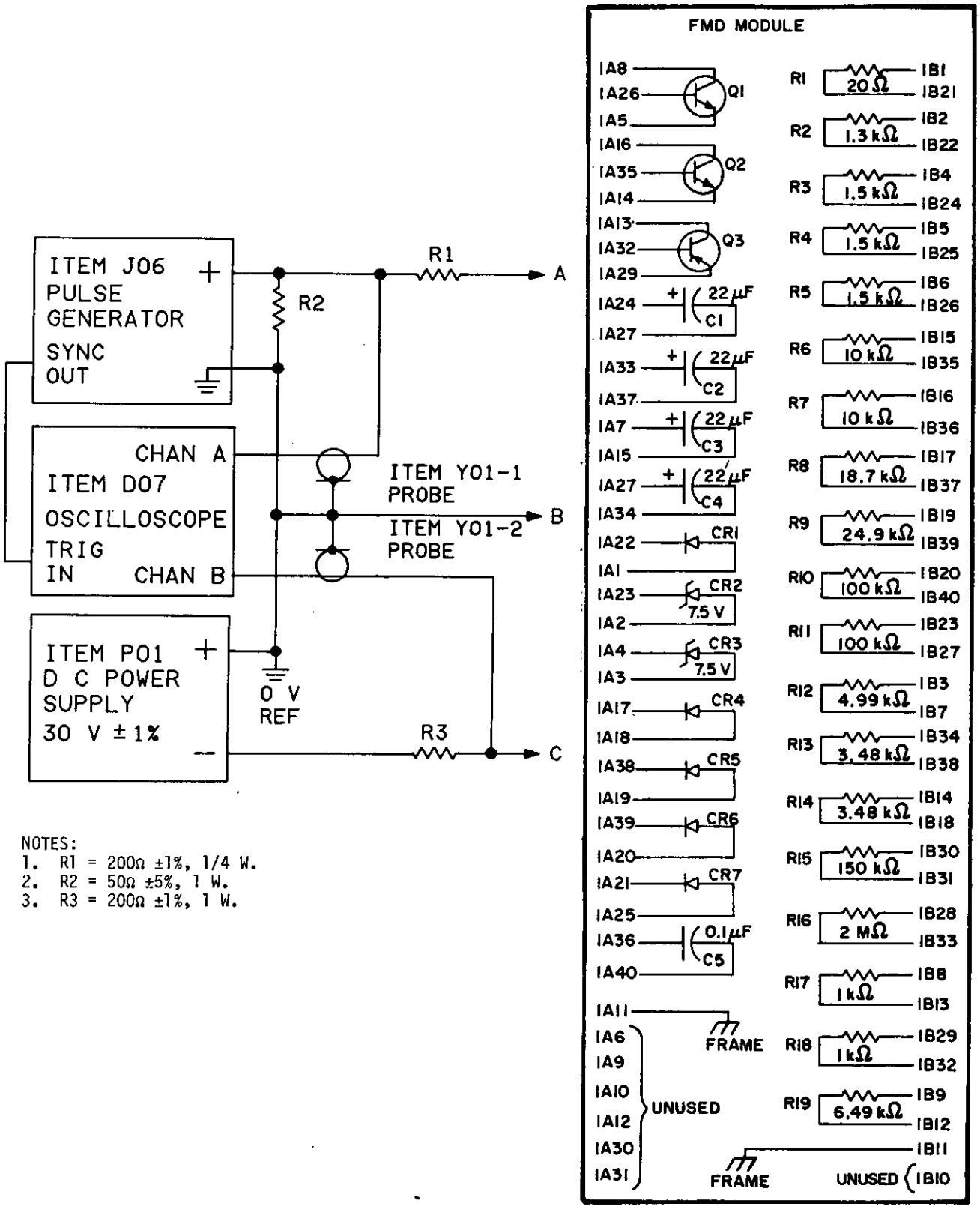
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NOTE: Input pulse parameters:
$t_{TLH} \leq 10$ ns.
PW = 200 ±10 ns.
PRP = 10.0 ±0.5 s.

* FIGURE 14. NPN turn-on time (t_{ON}) test circuit waveforms for table XII.

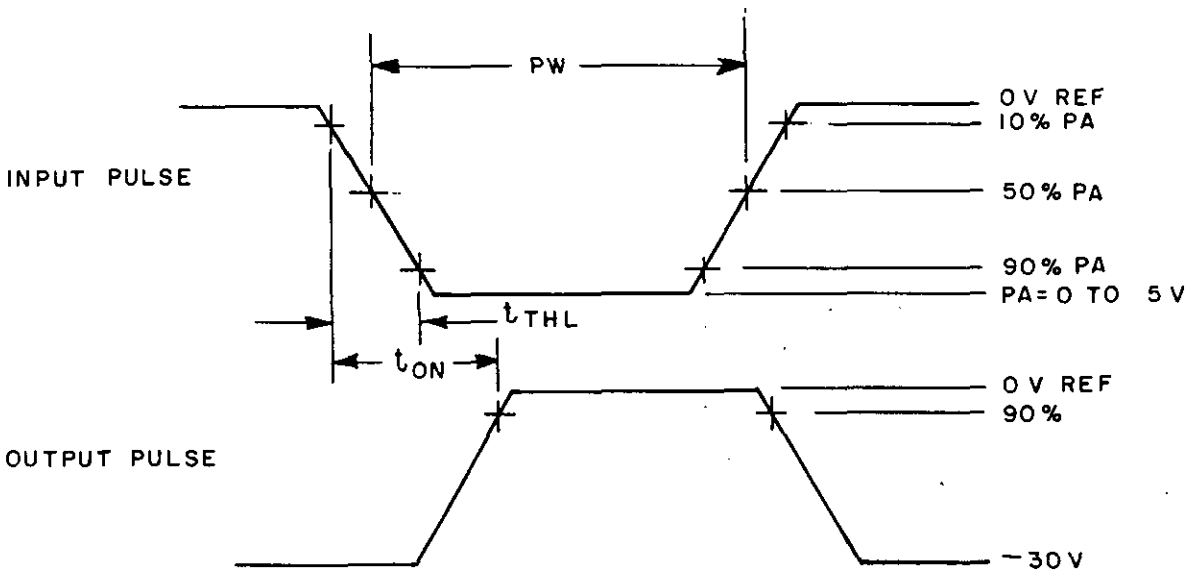
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NOTES:
1. R1 = 200Ω ±1%, 1/4 W.
2. R2 = 50Ω ±5%, 1 W.
3. R3 = 200Ω ±1%, 1 W.

FIGURE 15. PNP turn-on time (t_{ON}) test circuit for table XIII.

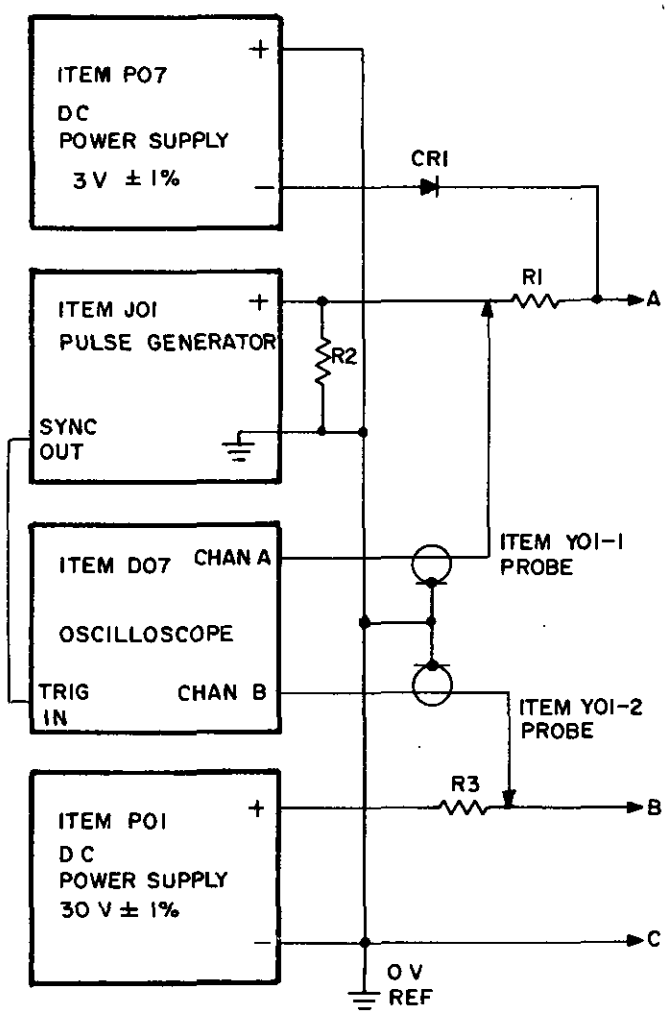
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NOTE: Input pulse parameters:
$t_{TLH} \leq 10$ ns.
PW = 200 ±10 ns.
PRP = 10.0±0.5 s.

* FIGURE 16. PNP turn-on time (t_{ON}) test circuit waveforms for table XIII.

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NOTES:
1. R1 = 340Ω ±1%, 1/2 W.
2. R2 = 50Ω ±5%, 1 W.
3. R3 = 200Ω ±1%, 1 W.
4. CR1 = 1N916 or equivalent.

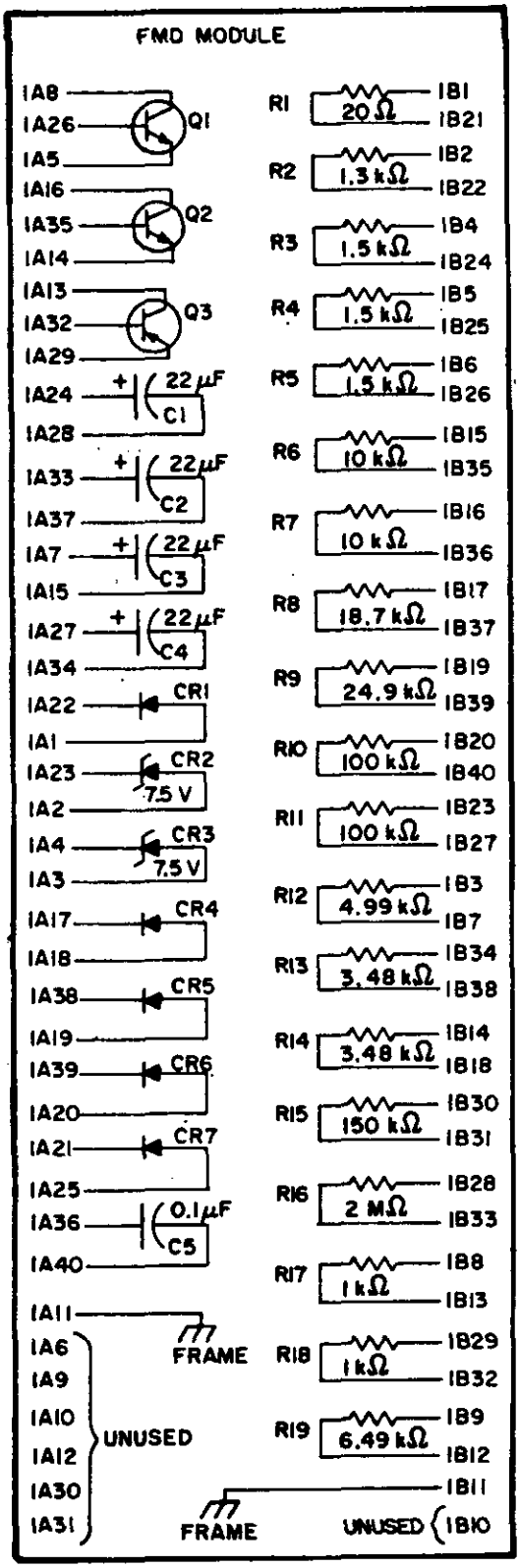
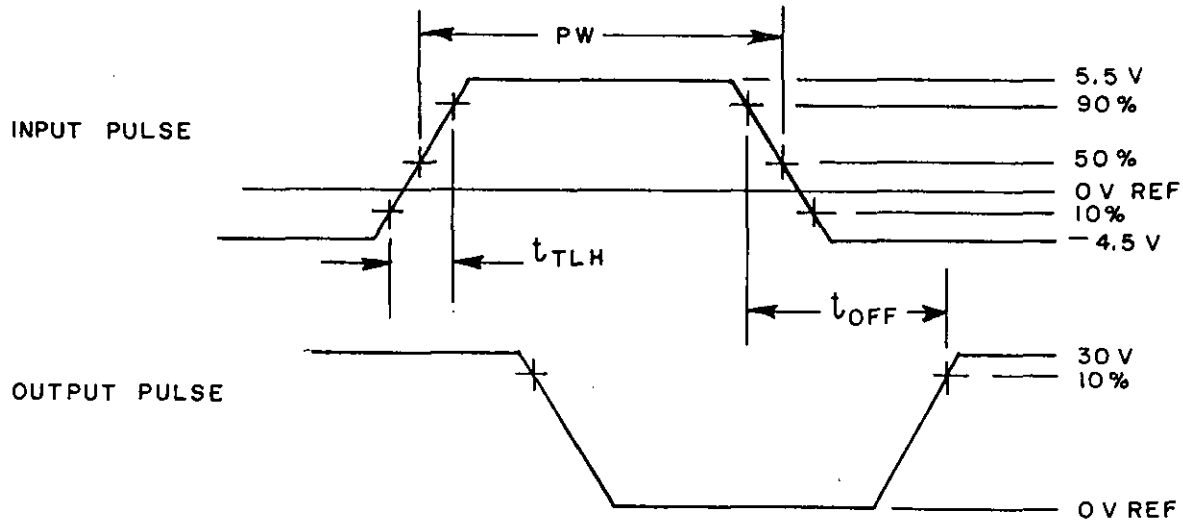


FIGURE 17. NPN turn-off time (t_{OFF}) test circuit for table XIV.

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NOTE: Input pulse parameters:
$t_{TLH} \leq 5$ ns.
PW = 20 ±1 µs
PRP = 1.0 ±0.1 ms.

FIGURE 18. NPN turn-off time (t_{OFF}) test circuit waveforms for table XIV.

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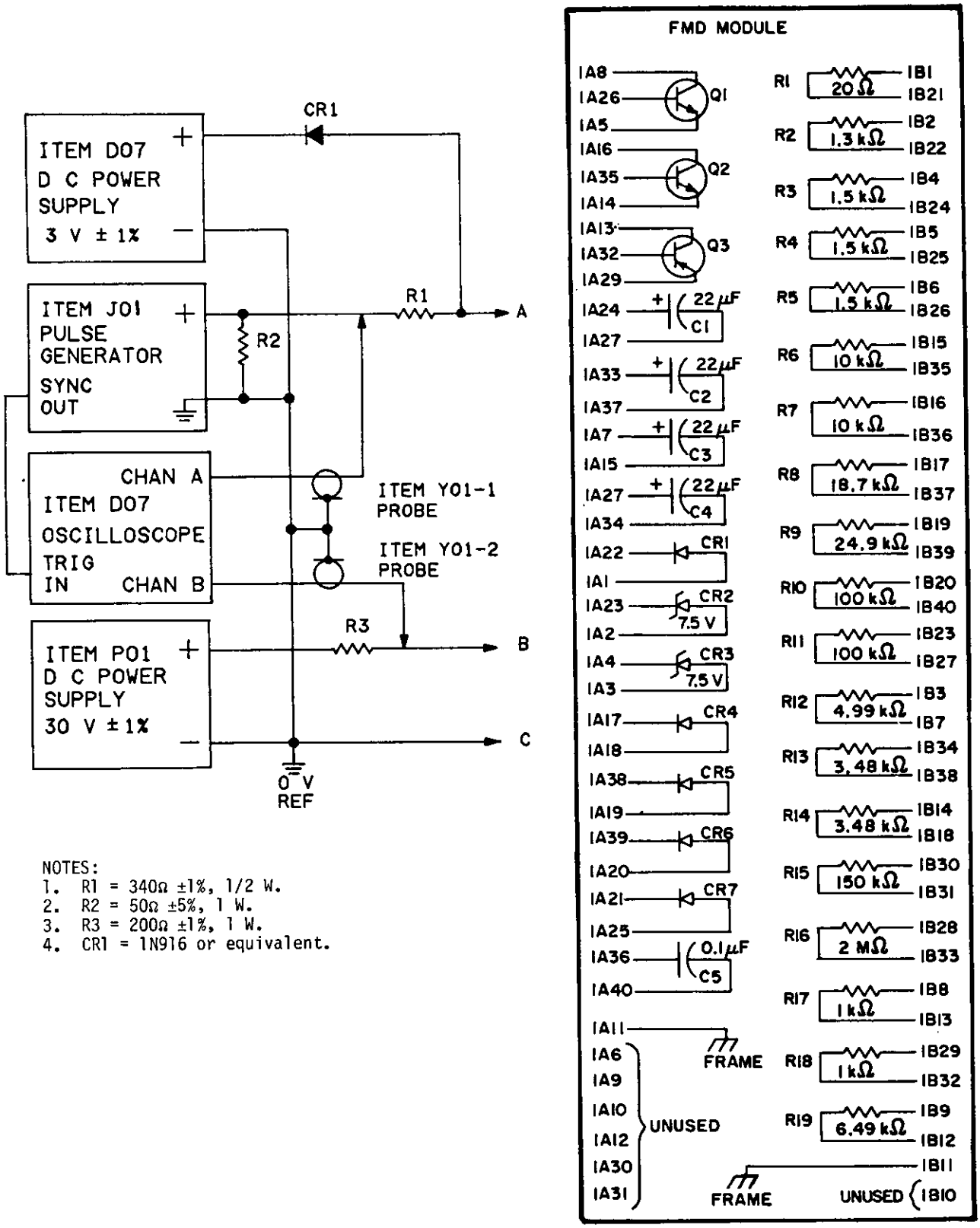
TABLE XV. PNP turn-off time (t_{OFF}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P3	Q3 PNP turn-off time (t_{OFF})	1A32	1A13	1A29	0	300	ns	19,20

TABLE XVI. Breakdown voltage check (emitter-to-base) test.

Test condition	Parameter	BV_{EBO}	I_E	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	Min	Max		
Q1	Q1 Breakdown voltage check (emitter-to-base)	6 V	10 µA	1A5	1A26	0	10	µA	21
Q2	"	6 V	"	1A14	1A35	"	"	"	"
Q3	"	5 V	"	1A32	1A29	"	"	"	"

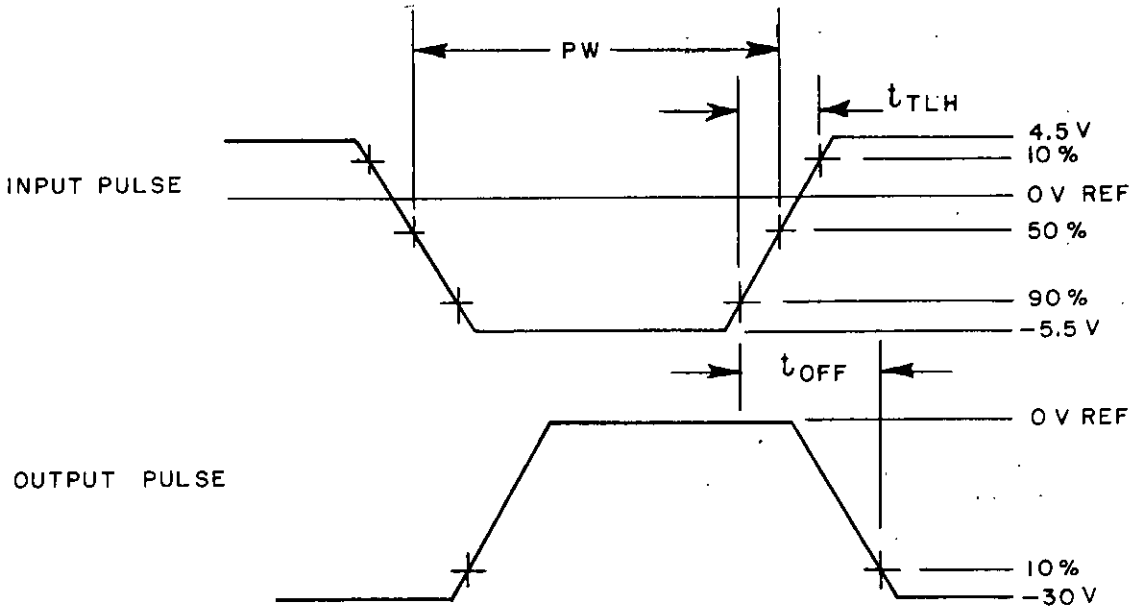
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NOTES:
1. R1 = 340Ω ±1%, 1/2 W.
2. R2 = 50Ω ±5%, 1 W.
3. R3 = 200Ω ±1%, 1 W.
4. CR1 = 1N916 or equivalent.

FIGURE 19. PNP turn-off time (t_{OFF}) test circuit for table XV.

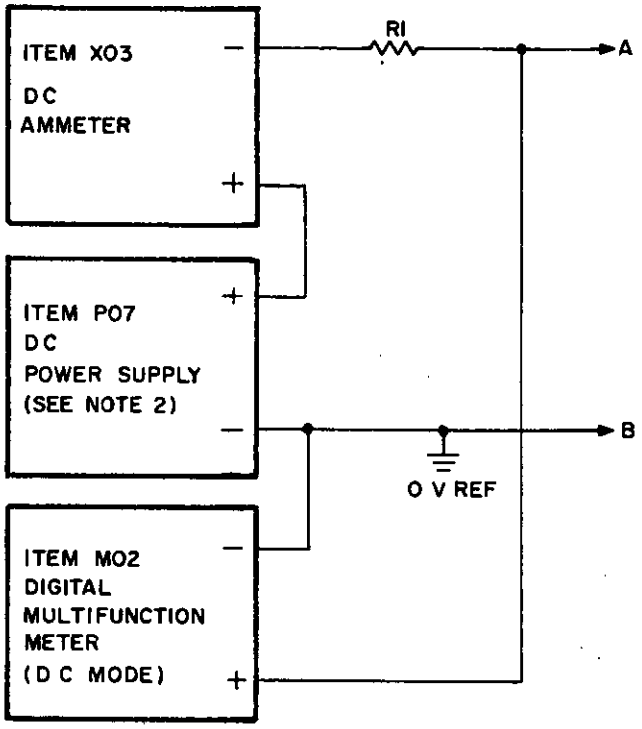
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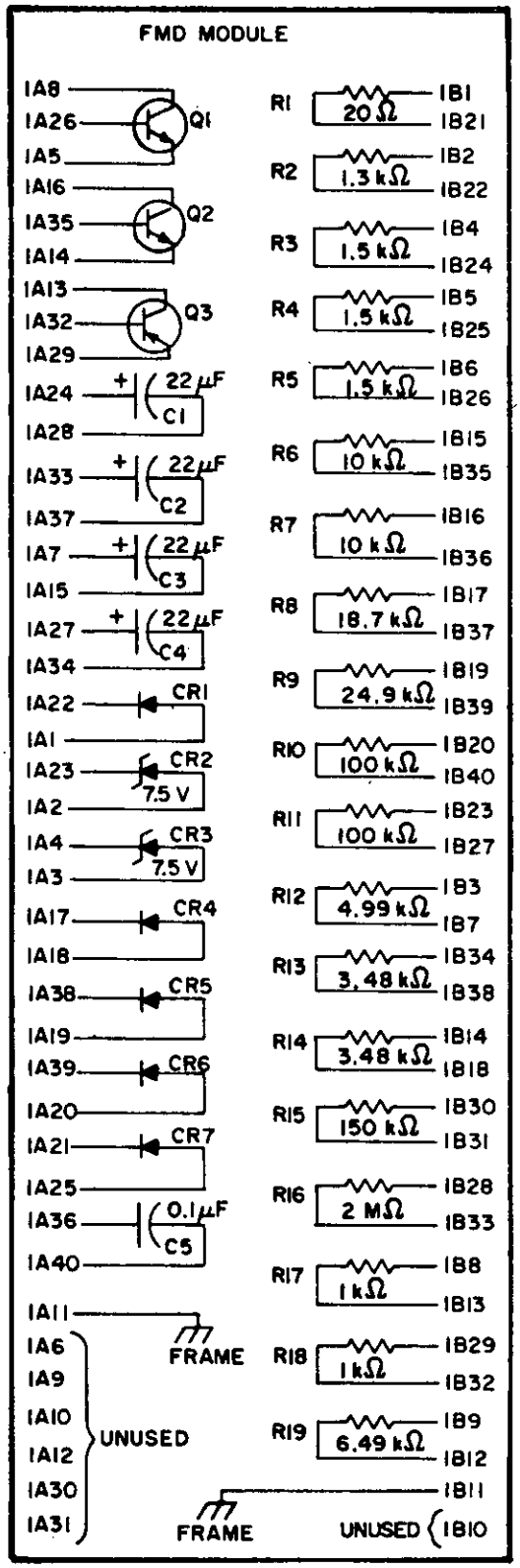


NOTE: Input pulse parameters:
$t_{TLH} \leq 5$ ns.
PW = 20 ±1 µs.
PRP = 1.0 ±0.1 ms.

FIGURE 20. PNP turn-off time (t_{OFF}) test circuit waveforms for table XV.

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NOTES:
1. R1 = 500 kΩ ±1%, 1/8 W.
2. The test shall be performed as follows:
   a. Initially set item PO7 to 0 V.
   b. Gradually increase the voltage on item PO7 until either the specified test current I_E is reached or the breakdown voltage BV_{EBO} is reached (Do not exceed the rated BV_{EBO}).
   c. The module passes if BV_{EBO} is reached before the specified I_E is reached (Do not exceed the maximum allowable limit for I_E). Record the item XO3 reading for each test condition.

FIGURE 21. Breakdown voltage check (emitter-to-base) test circuit for table XVI.

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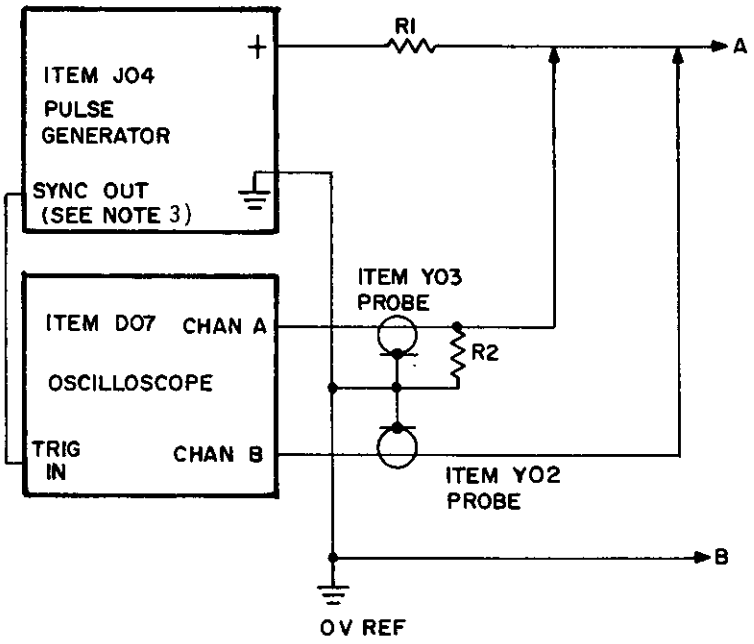
TABLE XVII. Breakdown voltage check (collector-to-emitter) test.

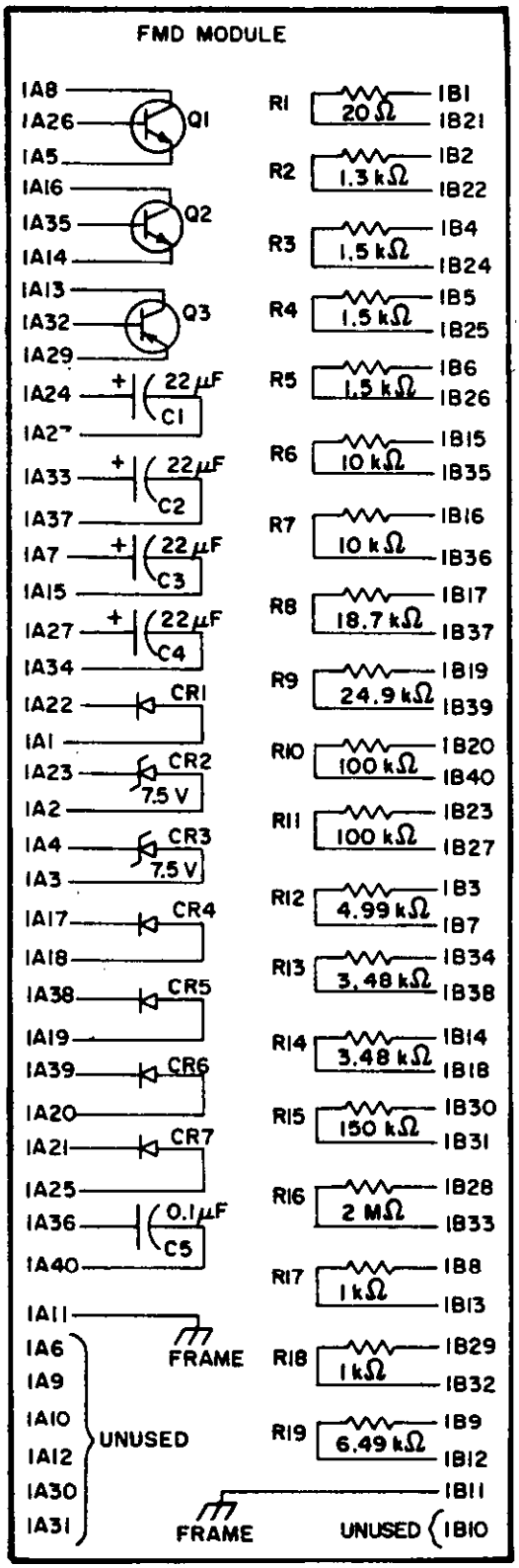
Test condition	Parameter	BV_{CEO}	I_C	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	Min	Max		
R1	Q1 Breakdown voltage check (collector-to-emitter)	50 V	10 mA	1A8	1A5	0	10	mA	22
R2	"	50 V	10 mA	1A16	1A14	"	10	"	"
R3	"	-60 V	-10 mA	1A13	1A29	"	-10	"	"

TABLE XVIII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S1	Isolation, frame	1A11	All other pins except 1B11	10	∞	MΩ	23
S2	Isolation, frame	1B11	All other pins except 1A11	"	"	"	"
S3	Isolation, used to unused pins	1A6,1A9,1A10, 1A12, 1A30, 1A31, 1B10	All other pins	"	"	"	"

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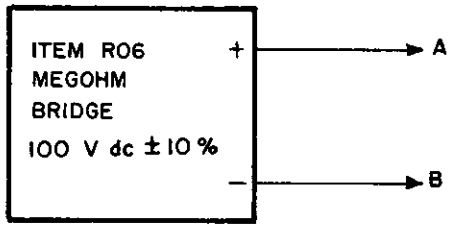




NOTES:
1. R1 = 1 kΩ ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.
3. Item J04 setup:
   a. PW = 20 ±2 µs.
   b. PRR = 1 kHz ±100 Hz.
   c. $t_{TLH} = t_{THL} \leq 2.5$ µs.
   d. For test conditions R1 and R2 set the polarity with baseline at 0 V.
   e. For test condition R3, set the polarity to negative with baseline at 0 V.
4. The test shall be performed as follows:
   a. Initially set item J04 amplitude to 0 V.
   b. Gradually increase the amplitude of item J04 until either the specified test current I_C is reached (read amplitude of pulse on channel A) or the breakdown voltage BV_{CEO} is reached (read amplitude of pulse on channel B). Do not exceed the indicated limits of I_C or BV_{CEO}.
   c. The module passes if BV_{CEO} is reached before the specified I_C is reached. Record the value of I_C measured on item D07. Measure current (I_C) 10 ±1 µs from 50% mark of rising edge of current waveform.

* FIGURE 22. Breakdown voltage check (collector-to-emitter) test circuit for table XVII.

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

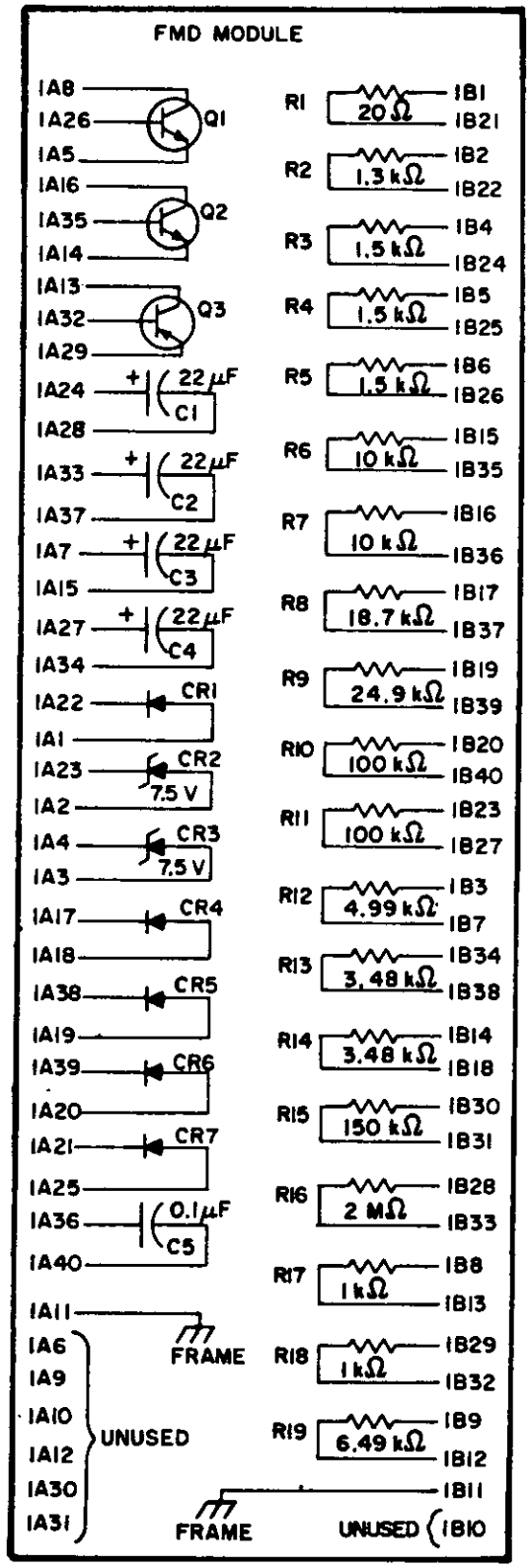


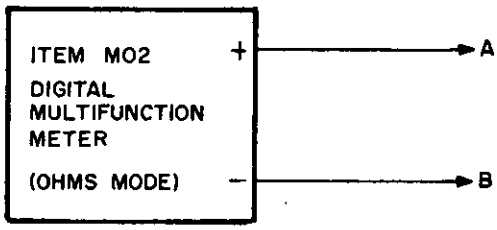
FIGURE 23. Isolation test circuit for table XVIII.

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TABLE XIX. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	Continuity, pin to frame	1A11	Frame	0	1	Ω	24
T2	Continuity, pin to frame	1B11	Frame	0	1	Ω	24

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NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe attached to the negative terminal.

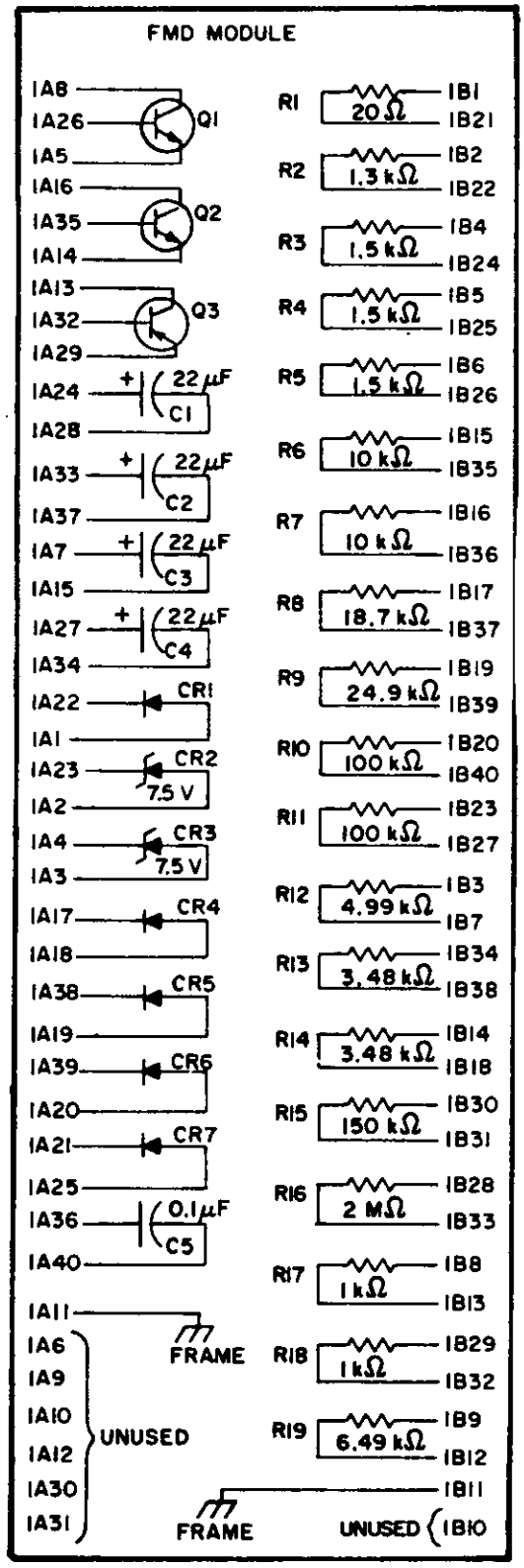


FIGURE 24. Continuity test circuit for table XIX.

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